5G RAN

Massive MIMO Uplink Boosting (Low-Frequency TDD) Feature Parameter Description

Issue Draft B

Date 2026-01-30

HUAWEI TECHNOLOGIES CO., LTD.

Huawei Technologies Co., Ltd.

Address:	Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China
Website:	https://www.huawei.com
Email:	support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:
https://www.huawei.com/en/psirt/vul-response-process
For vulnerability information, enterprise customers can visit the following web page:
https://securitybulletin.huawei.com/enterprise/en/security-advisory

Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes
  - Deleted the descriptions of the **STR NRCELLSERVICEPARAMEFFECT** command from **5.1.1.1 PUSCH Coordinated Power Control** and **5.4.1.2 Using MML Commands**.
  - Changed the value of the **NRCellMeasConfig**.***UlCpcUeSsbRsrpThld*** parameter in the MML command example. For details, see **5.4.1.2 Using MML Commands**.
  - Added descriptions of the impact of multi-cell power control for interference reduction and time-frequency scheduling for interference avoidance on the CPU usage. For details, see **5.2.2 Impacts**.
  - Added descriptions of the dependency of probability-based uplink MU-MIMO MCS selection on PUSCH MU-MIMO. For details, see **5.3.2.1 Prerequisite Functions**.
  - Corrected the value of the **NRDUCellUlSch**.***AdaptAwareSchTimeThld*** parameter in the MML command example. For details, see **5.4.1.2 Using MML Commands**.
  - Revised the descriptions of the principles of adaptive awareness scheduling. For details, see **5.1.2.3 Adaptive Awareness Scheduling**.
  - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the Uplink Boosting 2.0 feature. For details, see **5 Uplink Boosting 2.0**.	Added parameters: • **NRCellMeasConfig**.***UlCpcUeSsbRsrpThld*** • **NRDUCellUlPcConfig**.***UlCpcUeSrsRsrpThld*** • **NRDUCellUlPcConfig**.***UlCpcUeIntrfRsrpThld*** • **NRDUCellAiAlgo**.***UlMcsPredictErrUpLimit*** • **NRDUCellUlSch**.***AdaptAwareSchTimeThld*** • **NRDUCellUlSch**.***AdaptAwareSchPriFactor*** • **NRDUCellUlPcConfig**.***LinkPerfPcRsrpThldOffset*** • **NRDUCellUlSch**.***UlRetransSchDelayThld*** • **NRDUCellUlSch**.***PreSchPeriodAdjMcsThld*** Modified parameters: • Added the **UL_LOW_NOISE_PHASE2_SW** option to the **NRDUCellFeatureSw**.***MimoFeatureSwitch*** parameter. • Added the **PUSCH_PUCCH_RES_SHARE_SW** option to the **NRDUCellPusch**.***PuschAlgoExtSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
	• Added the **UL_RETRANS_PREC_RB_CTRL_SW** option to the **NRDUCellPusch**.***PuschAlgoExtSwitch*** parameter. • Added the **PUSCH_COORD_PWR_CTRL_SW** option to the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch*** parameter. • Added the **INTRF_PREC_FREQ_OFS_EST_SW** option to the **NRDUCellPusch**.***PuschMeasSw*** parameter. • Added the **UL_PRECISE_MCS_OPT_SW** option to the **NRDUCellAiAlgo**.***AiAmcAlgoSwitch*** parameter. • Added the **ADAPT_AWARE_SCH_SW** option to the **NRDUCellUlSch**.***UlSchAlgoSwitch*** parameter. • Added the **UL_MU_PBMCSSEL_SW** option to the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter. • Added the **INTRF_SC_LINK_PERF_PC_SW** option to the		

Change Description	Parameter Change	RAT	Base Station Model
	NRDUCellUlPcConfig.*UlPwrCtrlAlgoExtSwitch* parameter. • Added the **UL_EXP_SCH_OPT_SW** option to the **NRDUCellUlSch**.*UlSchAlgoSwitch* parameter. • Added the **PREC_CHANNEL_EST_SW** option to the **NRDUCellPusch**.*UlPuschAlgoSwitch* parameter. • Added the **MULTI_BEAM_RX_ENH_SW** option to the **NRDUCellPusch**.*PuschAlgoExtSwitch* parameter. Added the value **UL_PRECISE_MCS_OPT(UL Precise MCS Optimization)** to the **AiCase** parameter used in the **DSP NRDUCELLAISCHMODEL** command.		
Added data-volume-based intelligent power control optimization. For details, see: • 4.1.1.2 Scheduling Optimization • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added the **DATA_VOL_INTEL_PWR_CTRL_OPT_SW** option to the **NRDUCellUlPcConfig**.*UlPwrCtrlAlgoExtSwitch* parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between packet-length-based scheduling optimization and adaptive awareness scheduling. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between adaptive outer-loop adjustment for small-packet services and uplink precise MCS optimization. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added uplink MU grouping and pairing avoidance for RedCap UEs. For details, see: • **4.1.1.1 MU Pairing Optimization** • **4.3.2.1 Prerequisite Functions** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Modified parameter: Added the **REDCAP_UL_MU_GRP_PAIR_AVOID_SW** option to the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
  - Deleted the descriptions of the mutually exclusive relationships of PDCCH blind detection assignment optimization with the following functions from **4.3.2.2 Mutually Exclusive Functions**: super uplink, flexible super uplink, inter-base-station super uplink, and smart PDCCH symbol allocation.
  - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-091201	Uplink Boosting	4 Uplink Boosting
FOFD-100201	Uplink Boosting 2.0	5 Uplink Boosting 2.0

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Uplink Boosting	None	4 Uplink Boosting
Uplink Boosting 2.0	None	5 Uplink Boosting 2.0

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Uplink Boosting	Supported only in low frequency bands	4 Uplink Boosting
Uplink Boosting 2.0	Supported only in low frequency bands	5 Uplink Boosting 2.0

3 Overview

The growing volume of 5G traffic results in increased uplink loads and heightened intra-RAT interference. As a consequence, both traffic and control channels face resource shortages due to this interference. To address these challenges, Uplink Boosting **enhances uplink user experience in interference-prone environments**. Designed for TDD massive MIMO cells, this feature reduces interference on traffic channels, control channels, and reference signals, improving the overall uplink user experience in cells under interference.

With the rapid rise in both the number of 5G UEs and traffic volume, uplink loads are surging, leading to a shortage of scheduling resources across cells on the live network. This significantly increases inter-cell interference. Uplink Boosting 2.0 addresses these challenges for TDD massive MIMO cells through multi-cell power control for interference reduction, time-frequency scheduling for interference avoidance, and multi-domain demodulation for interference resistance. These measures boost uplink cell performance, **delivering an enhanced uplink user experience in contiguously covered areas experiencing interference and medium-to-high traffic loads**.

4 Uplink Boosting

4.1 Principles

As the number of 5G UEs increases rapidly, uplink loads in NR cells and intra-RAT interference gradually increase. As a result, uplink user experience deteriorates. Uplink Boosting is designed to address this issue. **Uplink Boosting improves uplink performance of TDD massive MIMO cells by reducing interference on traffic channels, control channels, and reference signals, ultimately increasing the average uplink UE throughput.** This function is controlled by the **UL_LOW_NOISE_SW** option of the **NRDUCellFeatureSw**.***MimoFeatureSwitch*** parameter.

4.1.1 Traffic Channel Interference Reduction

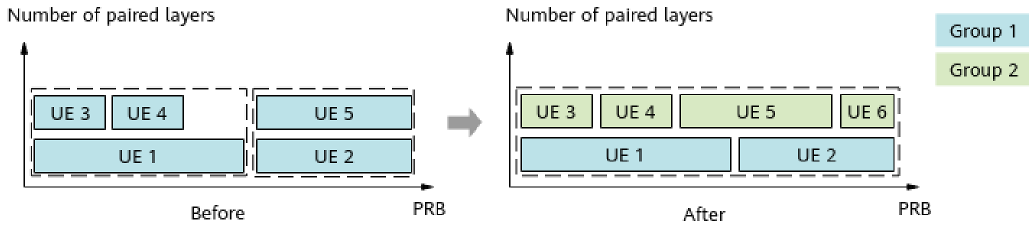
4.1.1.1 MU Pairing Optimization

Uplink MU Grouping and Pairing

Conventional uplink MU pairing, which is priority-based, preferentially pairs high-priority UEs. However, the overall pairing effect can still be improved in cells. To address this issue, uplink MU grouping and pairing is introduced.

Uplink MU grouping and pairing enables UEs to be grouped and paired in a way that preferentially guarantees the overall cell performance. It achieves this by considering factors such as UE distribution, services, and the number of layers. This function improves the overall pairing performance and average uplink UE throughput in a cell. As shown in **Figure 4-1**, this function allows more flexible pairing for UE 5 and allows UE 6 to participate in pairing, increasing the numbers of uplink paired PRBs and paired UEs in the cell.

Figure 4-1 Uplink MU grouping and pairing



This function is controlled by the **UL_MU_GRP_PAIR_SW** option of the **NRDUCellUlMimo**.***UlMuMimoAlgoSwitch*** parameter and takes effect when the number of uplink RBs required by all to-be-scheduled UEs in a cell divided by the number of available PUSCH RBs is greater than the threshold specified by the **NRDUCellUlMimo**.***FlexPairRbRatioThld*** parameter. With uplink MU grouping and pairing enabled, a minimum percentage of RBs is guaranteed for UEs involved in SU-MIMO scheduling. This minimum percentage is specified by the **NRDUCellUlMimo**.***MuGrpPairNormSuMinRbRatio*** parameter.

If there are RedCap UEs on the network, it is recommended that the following functions also be enabled:

- Uplink MU grouping and pairing optimization for RedCap UEs: This function optimizes the uplink MU grouping and pairing procedure for RedCap UEs, increasing the average uplink throughput of RedCap UEs.

  This function is controlled by the **REDCAP_UL_MU_GRP_PAIR_OPT_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter.
- Uplink MU grouping and pairing avoidance for RedCap UEs: In uplink scheduling, this function enables non-RedCap UEs with higher priorities to avoid RedCap UEs. This increases the average uplink throughput of RedCap UEs.

  This function is controlled by the **REDCAP_UL_MU_GRP_PAIR_AVOID_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter and takes effect only when the **REDCAP_UL_MU_GRP_PAIR_OPT_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter is selected.

For details about RedCap UEs, see the "RedCap Introduction" chapter in *RedCap*.

Mixed Pairing of UEs Using Different Waveforms

While UEs using different waveforms (CP-OFDM and DFT-s-OFDM) co-exist on the live network, only those using the same waveform can be paired, affecting MU pairing performance. (CP-OFDM is short for Cyclic Prefix-Orthogonal Frequency Division Multiplexing, and DFT-s-OFDM is short for Discrete Fourier Transform-spread OFDM.) To address this issue, mixed pairing of UEs using different waveforms is introduced.

This function allows UEs with different waveforms to be paired with each other, further improving pairing performance in cells and increasing the average uplink UE throughput.

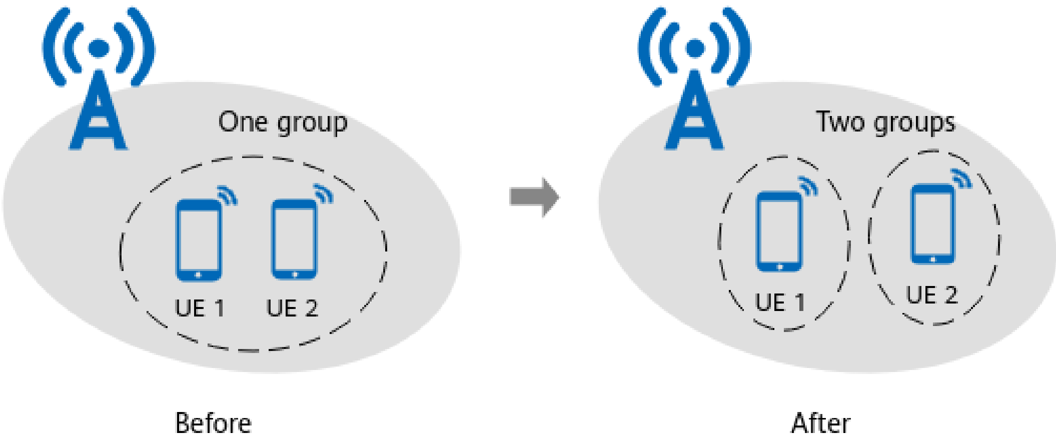
This function is controlled by the **DIFF_WAVEFORM_PAIR_SW** option of the **NRDUCellUlMimo**.***UlMuMimoAlgoSwitch*** parameter.

Optimized Uplink Correlation-based Pairing

When UEs are densely distributed in a cell, the base station cannot calculate the correlation between all UEs in a timely manner. As a result, some UEs cannot be paired, affecting the pairing performance of the cell. To address this issue, optimized uplink correlation-based pairing is introduced.

This function enhances the capability of calculating correlation between UEs, and enables high-resolution pairing in scenarios where UEs are concentrated. It increases the proportion of uplink paired UEs in a cell and the average uplink cell throughput. As shown in **Figure 4-2**, this function sets apart UEs originally in the same group into different groups for correlation calculation, increasing the number of uplink paired UEs and the proportion of uplink paired UEs in the cell.

Figure 4-2 Optimized uplink correlation-based pairing



This function is controlled by the **UL_CORR_ACCELERATION_SW** option of the **NRDUCellUlMimo**.***UlMuMimoAlgoSwitch*** parameter.

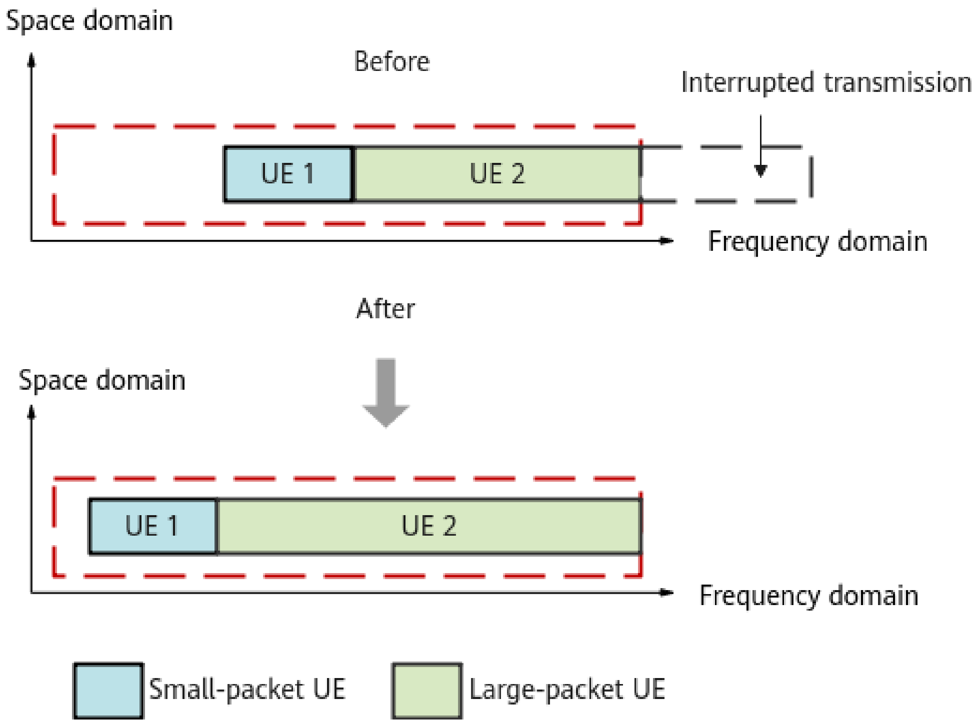
4.1.1.2 Scheduling Optimization

Adaptive PUSCH Resource Allocation

On the live network, frequency selective scheduling of small-packet UEs that have high scheduling priorities would interrupt the scheduling of large-packet UEs. To address this issue, adaptive PUSCH resource allocation has been introduced.

This function adaptively selects a frequency selective scheduling policy for small-packet UEs, reducing the probability for small-packet UEs to interrupt the scheduling of large-packet UEs. This increases the average uplink cell throughput. As shown in **Figure 4-3**, the scheduling of UE 2 (a large-packet UE) is no longer interrupted after this function is enabled so that more RBs are scheduled.

Figure 4-3 Adaptive PUSCH resource allocation



This function is controlled by the **PUSCH_RES_ADAPT_ALLOC_SW** option of the **NRDUCellPusch**.***PuschAlgoExtSwitch*** parameter.

Uplink Preallocation Period Adjustment

In NSA networking, if the preallocation period of a UE is short, the uplink padding rate of the cell and the uplink interference to neighboring cells increase. To address this issue, uplink preallocation period adjustment is introduced.

This function adaptively adjusts the minimum period for uplink preallocation based on the UE's traffic model, reducing the number of preallocations. This reduces the uplink padding rate of the cell and uplink interference to neighboring cells, and increases the average uplink UE throughput.

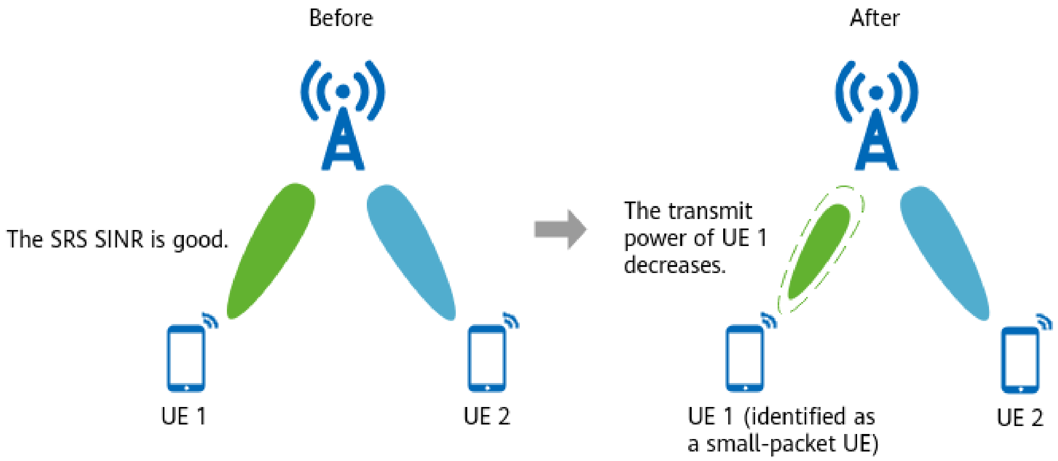
This function is controlled by the **UL_PREALLOCATION_PERIOD_ADJ_SW** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter.

Data-Volume-based Intelligent Power Control

On the live network, uplink services are mainly small-packet services. If the transmit power of small-packet UEs is too high, neighboring cells experience interference; it also leads to a power waste and higher power consumption. To address this issue, data-volume-based intelligent power control is introduced.

This function intelligently identifies the interference and traffic model of a cell and adaptively adjusts the transmit power and scheduling model for small-packet UEs that meet certain conditions in the cell. This reduces interference to neighboring cells and increases the average uplink UE throughput. As shown in **Figure 4-4**, this function reduces the transmit power of a qualified small-packet UE (UE 1).

Figure 4-4 Data-volume-based intelligent power control



This function is controlled by the **DATA_VOL_INTEL_PWR_CTRL_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch*** parameter. This function takes effect for a small-packet UE when its SRS SINR is greater than the threshold specified by the **NRDUCellUlPcConfig**.***SmallPktUeSrsSinrThld*** parameter. The amount of power reduced by does not exceed the value of the **NRDUCellUlPcConfig**.***SmallPktUePuschPwrOffset*** parameter.

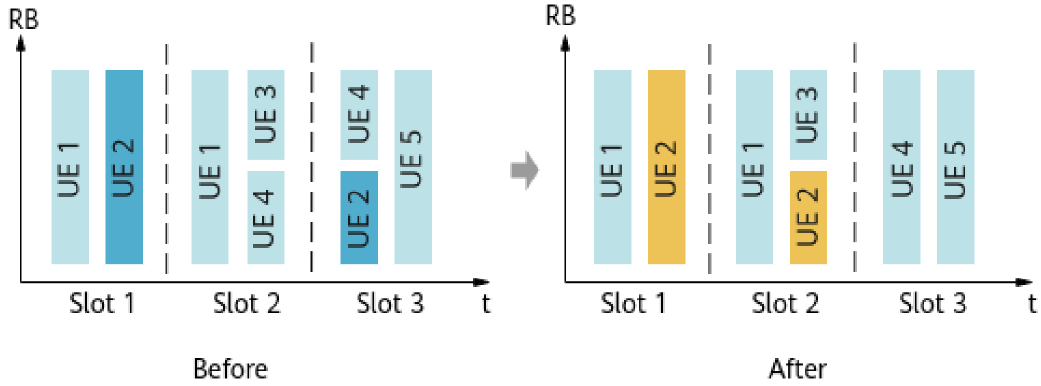
In certain cases, a large reduction in UE power can degrade user experience. It is therefore recommended that the **DATA_VOL_INTEL_PWR_CTRL_OPT_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch*** parameter be selected to enable data-volume-based intelligent power control optimization. This function enables the gNodeB to adaptively optimize the power adjustment amount based on UEs' channel quality to increase the average uplink UE throughput.

Packet-Length-based Scheduling Optimization

Since traffic models of UEs differ greatly on the live network, the total transmission delay of UEs can be reduced. Packet-length-based scheduling optimization is introduced to achieve this purpose.

Aimed at minimizing the total transmission delay of UEs to be scheduled, this function adaptively selects appropriate UEs for scheduling based on their traffic models and signal quality. This increases the average uplink UE throughput. Figure 4-5 shows how this function works. With this function, UE 2's data that used to be scheduled in slot 3 is now scheduled in slot 2, shortening the total scheduling duration of UE 2 by one slot.

Figure 4-5 Packet-length-based scheduling optimization



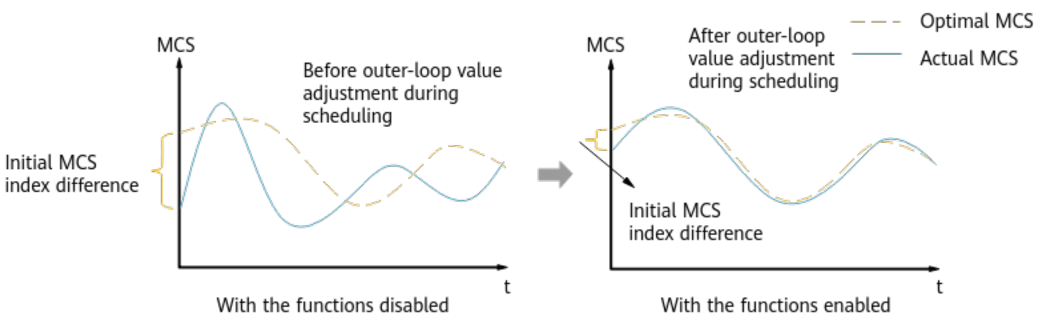
This function is controlled by the **PKT_LEN_BASED_SCH_OPT_SW** option of the **NRDUCellPusch**.***PuschPerformanceSwitch*** parameter.

Uplink Outer-Loop Optimization

Small-packet services account for a large proportion in the uplink on the live network, and the channel quality fluctuates greatly over the air interface. However, a fixed initial MCS index is used for all UEs. Since the demodulation performance of UEs varies with UE locations and scheduling types, it takes a long time for some UEs to reach the optimal MCS index through outer-loop adjustment, affecting user experience. To address this issue, uplink outer-loop optimization is introduced.

Uplink outer-loop optimization accelerates the process for a UE to reach the optimal MCS index. This is done through the following functions: (1) uplink cell-level outer-loop link adaptation (OLLA), which optimizes the initial OLLA value used by a UE; (2) adaptive outer-loop adjustment for small-packet services, which speeds up the MCS convergence process. Figure 4-6 shows how these functions work.

Figure 4-6 Uplink outer-loop optimization



- Uplink cell-level OLLA

  After uplink cell-level OLLA is enabled, the base station measures the average OLLA value of UEs at different locations (edge or center) of a cell in different slots. It then assigns an OLLA value for initial uplink MCS selection to an admitted UE based on its conditions so that the MCS

index can quickly get close to its optimal value. This function is controlled by the **UL_CELL_OLLA_SW** option of the **NRDUCellUlAmc**.***UlAmcAlgoSw*** parameter.

- Adaptive outer-loop adjustment for small-packet services

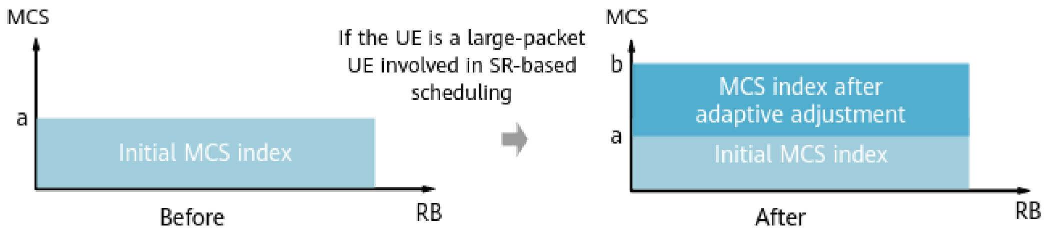
  After adaptive outer-loop adjustment for small-packet services is enabled, the base station adaptively adjusts the outer-loop values for small-packet services so that the MCS index can quickly get close to its optimal value for small-packet UEs. This increases the average uplink UE throughput. This function is controlled by the **SMALL_PKT_OL_ADAPT_ADJ_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter and takes effect for a UE when its SINR is greater than the threshold specified by the **NRDUCellUlAmc**.***OlAdaptAdjSinrThld*** parameter.

MCS Index Optimization for SR-based Scheduling

In SR-based scheduling, the MCS index of a UE is reduced to improve the reliability of demodulation performance. However, MCS index reduction of large-packet UEs decreases their average uplink UE throughput. To address this issue, MCS index optimization for SR-based scheduling is introduced.

This function adaptively selects MCS indexes for large-packet UEs based on their traffic volumes. As shown in **Figure 4-7**, this function adaptively selects an MCS index for a large-packet UE using a reduced MCS index for SR-scheduling.

Figure 4-7 MCS index optimization for SR-based scheduling



This function is controlled by the **SR_BASED_SCH_MCS_OPT_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter.

NOTE

SR-based scheduling refers to the first data scheduling after the gNodeB receives the scheduling request indicator (SRI) from a UE. For details about the definition of SRI, see "Basic Procedure of Uplink Scheduling" in *Uplink Scheduling*.

DCI-based Dynamic Waveform Switching

Currently, UEs switch between different waveforms through RRC signaling. This may raise the service drop rate of UEs at the cell edge or at a medium distance from the cell center. To address this issue, DCI-based dynamic waveform switching has been introduced.

This function adaptively instructs a UE through DCI to select the CP-OFDM or DFT-s-OFDM waveform based on the channel conditions and the threshold

specified by the **NRDUCellPusch**.***SinrThldforWaveformSel*** parameter. **This reduces the number of RRC reconfiguration messages and the service drop rate. In addition, the transmit power of a UE can be increased when the UE switches from the CP-OFDM waveform to the DFT-s-OFDM waveform.**

This function is controlled by the **DYNAMIC_WAVEFORM_SWITCHING_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter. The DFT-s-OFDM waveform is selected for a UE if its uplink SINR is less than the preceding threshold and rank 1 is used, and the CP-OFDM waveform is selected in other cases.

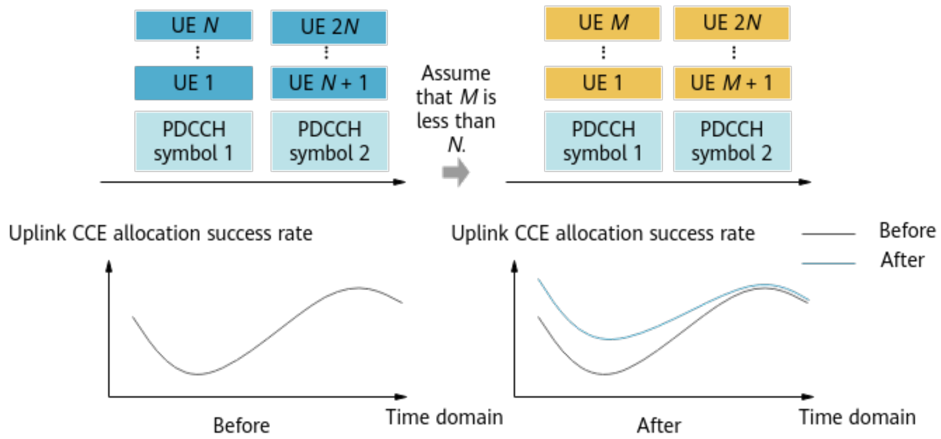
4.1.2 Control Channel Interference Reduction

Smart PDCCH Symbol Allocation

Currently, the CCE allocation policy is too slow to satisfy UEs' time-varying needs for CCEs. As a result, CCE resource allocation may fail, affecting user experience. To address this issue, smart PDCCH symbol allocation is introduced.

This function optimizes the policy for adjusting the number of available PDCCH symbols and the policy of CCE load balancing across symbols to more accurately match CCE requirements of UEs. This increases the uplink CCE allocation success rate and the average uplink UE throughput. As shown in **Figure 4-8**, UEs are no longer evenly distributed across PDCCH symbols after this function is enabled. Instead, UE distribution is dynamically adjusted based on the CCE requirements of UEs, increasing the uplink CCE allocation success rate for UEs.

Figure 4-8 Smart PDCCH symbol allocation



This function is enabled when the **PDCCH_SYM_SMART_ALLOC_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter is selected. When this function is enabled in certain scenarios, some functions also need to be enabled to ensure cell performance. Specifically:

- When this function is enabled in a cell with a bandwidth of 40 MHz to 70 MHz, CCE symbol allocation optimization (controlled by the **CCE_SYMBOL_ALLOC_OPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter) is required. This is to increase the uplink and downlink CCE allocation success rates in the cell and the average uplink and downlink UE throughputs.

- If the assignment of blind detections to each PDCCH aggregation level is uneven, uplink and downlink CCE allocation may fail due to insufficient blind detections. To balance the assignment of blind detections across different PDCCH aggregation levels, PDCCH blind detection assignment optimization (controlled by the **PDCCH_BLIND_DET_ASSIGN_OPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter) is required. This is to prevent uplink and downlink CCE allocation failures caused by insufficient blind detections.

Certain UEs are incompatible with smart PDCCH symbol allocation. To address this compatibility issue, it is recommended that blacklist control for dynamic PDCCH symbol allocation be enabled by selecting the **PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlExtSwitch*** parameter. This blacklist control function disables smart PDCCH symbol allocation for such UEs.

BWP2 CSI Period Optimization

With power saving BWP (BWP is short for bandwidth part) enabled, if the narrow-bandwidth BWP (called BWP2) is used, the RBs allocated to the PUCCH for periodic CSI reporting are limited. To address this issue, BWP2 CSI period optimization has been introduced.

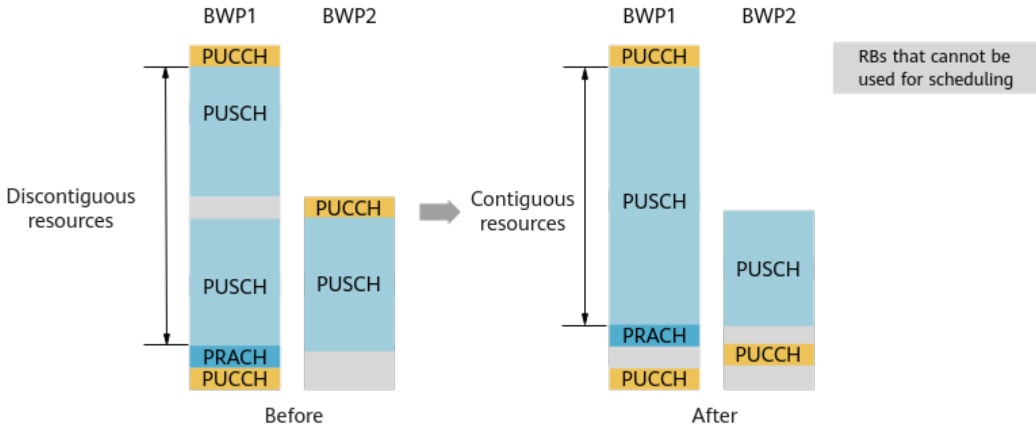
This function prolongs the PUCCH CSI period of BWP2 UEs to increase the number of UEs that can use a given number of PUCCH RBs. This prevents UEs from occupying PUSCH scheduling resources due to failures to obtain periodic PUCCH CSI resources, increasing the average uplink UE throughput. For details about power saving BWP and BWP2, see "Power Saving BWP (Low-Frequency TDD)" in *UE Power Saving*. This function is controlled by the **BWP2_CSI_PERIOD_OPT_SWITCH** option of the **NRDUCellPucch**.***CsiResoureAlgoSwitch*** parameter.

BWP2 PUCCH Frequency-Domain Position Optimization

After the power saving BWP function is enabled, the PUSCH resources in BWP1 may be truncated by the PUCCH resources in BWP2. This results in a decrease in the number of consecutive available RBs for the PUSCH and fragmentation of scheduling resources. As a result, the average uplink UE throughput is affected. To address this issue, the BWP2 PUCCH frequency-domain position optimization function is implemented. For details about power saving BWP, see *UE Power Saving*.

This function relocates the BWP2 PUCCH in the frequency domain to allow for contiguous PUSCH positions, which leads to a larger number of consecutive PUSCH RBs for the BWP1, as shown in **Figure 4-9**. **This increases the average uplink UE throughput.**

Figure 4-9 Example of BWP2 PUCCH frequency-domain position optimization



This function is controlled by the **BWP2_PUCCH_POS_OPT_SW** option of the **NRDUCellPucch.*PucchAlgoSwitch*** parameter.

When this function is used together with RedCap Introduction (controlled by the **REDCAP_SW** option of the **NRDUCellFeatureSw.*UeCapbBasedFunSw*** parameter), the frequency-domain position of the BWP2 PUCCH is different from that when this function is solely used. For the descriptions and network impacts of using this function together with RedCap Introduction, see the descriptions of BWP2 PUCCH frequency-domain position optimization in *RedCap*.

PUCCH F1 ACK Code Channel Interference Avoidance

An increased number of downlink scheduling times in an NR cell leads to more ACKs/NACKs reported by UEs. As a result, the interference between PUCCH code channels increases. To cope with this issue, PUCCH F1 ACK code channel interference avoidance has been introduced (F1 ACK is short for format-1 HARQ-ACK).

This function preferentially staggers PUCCH F1 ACK resources in the frequency domain before allocation to UEs. This reduces interference between F1 ACK code channels. For details about PUCCH format 1, see *Channel Management*.

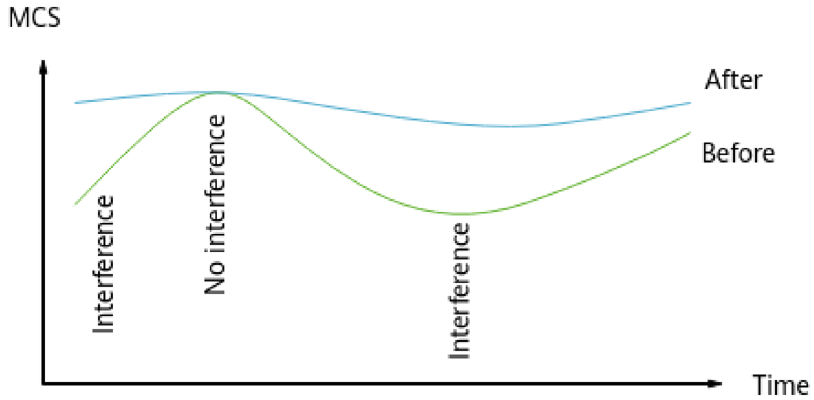
This function is controlled by the **F1_ACK_CODE_CHN_INTRF_OPT_SW** option of the **NRDUCellPucch.*PucchAlgoExtSwitch*** parameter.

4.1.3 Reference Signal Interference Reduction

When UEs at the cell edge experience interference, DMRS measurement is inaccurate. As a result, PUSCH demodulation performance deteriorates, affecting uplink user experience. To address this issue, IRC-based PDP detection has been introduced. (IRC is short for interference rejection combining and PDP is short for power delay profile.)

This function enables PDP detection after IRC is performed, improving the accuracy of PDP detection. This enhances the DMRS anti-interference capability of cell-edge UEs, improves their uplink demodulation performance, and increases the average uplink UE throughput. As shown in Figure 4-10, this function significantly increases the MCS indexes of UEs experiencing interference.

Figure 4-10 IRC-based PDP detection



This function is controlled by the **IRC_BASED_PDP_DETECT_SW** option of the **NRDUCellDmrs**.***UlDmrsSwitch*** parameter.

4.2 Network Analysis

4.2.1 Benefits

Uplink Boosting increases the average uplink UE throughput. This function applies to cells providing contiguous coverage, and it delivers the optimal benefits when all the following conditions are met:

- The uplink PRB usage of a cell is greater than or equal to 30%.
- The average uplink interference of neighboring cells (**N.UL.NI.Avg**) is greater than or equal to –110 dBm.
- The uplink CCE allocation success rate of a cell is less than or equal to 95% in the case that the uplink PRB usage of the cell is less than 50%.

The specific benefits depend on factors such as the traffic model and network load. Benefits are greater with a higher uplink PRB usage, higher average uplink interference, higher uplink padding rate, and lower uplink CCE allocation success rate. When the cell load is light (for example, when the uplink PRB usage is less than 30%) or the average uplink interference (**N.UL.NI.Avg**) is less than –110 dBm, some sub-functions take effect in a smaller proportion of cases, due to which fewer or even no benefits will be obtained.

NOTE

Average uplink UE throughput = (**N.ThpVol.UL** – **N.ThpVol.UE.UL.SmallPkt**)/**N.ThpTime.UE.UL.RmvSmallPkt**

Uplink PRB usage of a cell = **N.PRB.UL.Used.Avg**/**N.PRB.UL.Avail.Avg** x 100%

Uplink padding rate of a cell = 1 – (**N.ThpVol.UL**/**N.ThpVol.UL.Cell**) x 100%

Uplink CCE allocation success rate of a cell = (**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num** x 100%

4.2.2 Impacts

Functions of all three categories may have the following impacts:

- Improved uplink performance decreases the overall online duration of UEs in a cell and increases the number of resynchronizations for UEs performing periodic services. This may decrease the contention-based access success rate of such UEs.
- If there are only UEs near the cell center or the traffic volume of tail packets accounts for a large proportion in a cell, improved uplink performance may increase the uplink cell throughput and spectral efficiency but may decrease the average uplink UE throughput.

The following table describes the additional impacts of functions of each category.

<table>
<tr><th>Category</th><th>Network Impact</th></tr>
<tr><td>Traffic channel interference reduction</td><td>

The number of uplink paired PRBs (N.ChMeas.MIMO.UL.Pair.PRB) and the number of layers (N.ChMeas.MIMO.UL.Pair.Layer) in the cell fluctuate.
The uplink data rate at the MAC layer in the cell may fluctuate.

As the transmit power of small-packet UEs decreases, the average uplink MCS index and uplink padding rate of the cell may decrease. As a result, the uplink data rate at the MAC layer in the cell decreases and the uplink PRB usage of the cell increases. The SRS RSRP and SINR as well as the PUSCH RSRP and SINR of the cell may also decrease. This will cause more UEs to fall back from NR to LTE for uplink data transmissions and the value of N.NsaDc.ULChangetoMeNB to increase.
In medium- and light-load scenarios, adaptive selection of a frequency selective scheduling policy for small-packet UEs may cause fewer occasions where the scheduling of large-packet UEs is interrupted, increasing the uplink data rate at the MAC layer in the cell.

The uplink BLER of the cell may fluctuate.

When the demodulation performance improves or the interference decreases in the cell, the uplink BLER of the cell may decrease.
When there are a large number of cell-edge UEs or the cell experiences strong narrowband interference, the uplink BLER of the cell may increase, and the service drop rate in the cell may increase in extreme scenarios.

The proportion of uplink rank 1 in the cell may slightly increase.
The CPU usage (VS.BBUBoard.CPULoad.Mean) may increase.
The average uplink cell throughput may decrease if both of the following conditions are met for the cell:

There are a large number of UEs near the cell center, and these UEs have bursts of large-packet services.
The proportion of overlapping coverage between this cell and its neighboring cells is low.

The number of RRC reconfiguration messages may decrease.

</td></tr>
</table>

Category	Network Impact
Control channel interference reduction	• The uplink and downlink CCE usages of a cell may change, and the uplink and downlink CCE aggregation levels may decrease. • The average CQI of a cell may fluctuate. • The downlink PRB usage of a cell may increase. • The duration in which BWP2 is in effect in a cell may change. • The proportion of DRX sleep time in a cell may fluctuate. • The number of RRC reconfiguration messages may decrease. • It is recommended that BWP2 PUCCH frequency-domain position optimization be enabled when multiple cells provide contiguous coverage. If this function is enabled in other scenarios, PUCCH interference between cells may change, which may cause value changes in indicators such as the PDCCH aggregation level, average uplink and downlink cell throughputs, and uplink and downlink BLERs.
Reference signal interference reduction	No impact

The impacts between Uplink Boosting and other functions are described in **4.3.2.3 Function Impacts**.

NOTE

Average uplink MCS index in a cell = SUM(*k* x **N.ChMeas.PUSCH.MCS.*k***)/SUM(**N.ChMeas.PUSCH.MCS.*k***), where *k* = 0 to 28

Uplink IBLER of a cell = (**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**)

Uplink data rate at the MAC layer in a cell = **N.ThpVol.UL.Cell**/**N.ThpTime.UL.Cell**

Proportion of uplink rank 1 in a cell = **N.PUSCH.TbUl.Rank1**/(**N.PUSCH.TbUl.Rank1** + **N.PUSCH.TbUl.Rank2**)

Service drop rate in a cell = **N.UECntx.AbnormRel**/(**N.UECntx.NormRel** + **N.UECntx.AbnormRel**) x 100%

Uplink CCE usage of a cell = **N.CCE.ULDCI.Used.Avg**/**N.CCE.Avail.Avg** x 100%

Downlink CCE usage of a cell = **N.CCE.DLDCI.Used.Avg**/**N.CCE.Avail.Avg** x 100%

Average uplink CCE aggregation level of a cell = (**N.CCE.UL.AggLvl16Num** x 16 + **N.CCE.UL.AggLvl8Num** x 8 + **N.CCE.UL.AggLvl4Num** x 4 + **N.CCE.UL.AggLvl2Num** x 2+ **N.CCE.UL.AggLvl1Num**)/(**N.CCE.UL.AggLvl16Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl1Num**)

Average downlink CCE aggregation level of a cell = (**N.CCE.DL.AggLvl16Num** x 16 + **N.CCE.DL.AggLvl8Num** x 8 + **N.CCE.DL.AggLvl4Num** x 4 + **N.CCE.DL.AggLvl2Num** x 2+ **N.CCE.DL.AggLvl1Num**)/(**N.CCE.DL.AggLvl16Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl1Num**)

Average CQI of a cell = SUM(*k* x **N.ChMeas.CQI.SingleCW.*k***)/SUM(**N.ChMeas.CQI.SingleCW.*k***), where *k* = 0 to 15

Downlink PRB usage of a cell = **N.PRB.DL.Used.Avg**/**N.PRB.DL.Avail.Avg** x 100%

Duration in which BWP2 is in effect in a cell = **N.Bwp2.Dur.Total**

Proportion of DRX sleep time in a cell = **N.Cdrx.Sleep.Dur.Total**/(**N.Cdrx.Sleep.Dur.Total** + **N.Cdrx.Active.Dur.Total**) x 100%

Contention-based access success rate = **N.RA.Contention.Resolution.Succ**/**N.RA.Contention.Att** x 100%

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-091201	Uplink Boosting	NR0S00UAHR00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

Functions of traffic channel interference reduction require the following functions.

Function Name	Function Switch	Reference	Description
PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	Optimized uplink correlation-based pairing can take effect only when PUSCH MU-MIMO is enabled.
Uplink scheduling enhancement	**UL_SCHEDULE_ENH_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	*MIMO (TDD)*	Optimized uplink correlation-based pairing can take effect only when uplink scheduling enhancement is enabled.
Uplink DMRS Type	**NRDUCellPusch**.***UlDmrsType*** set to **TYPE1**	*Uplink Scheduling*	Uplink MU grouping and pairing can be enabled only when uplink DMRS type 1 is used.
PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	Uplink MU grouping and pairing can take effect only when PUSCH MU-MIMO is enabled.
Uplink MU grouping and pairing	**UL_MU_GRP_PAIR_SW** option of the **NRDUCellUlMimo**.***UlMuMimoAlgoSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	Mixed pairing of UEs using different waveforms can take effect only when uplink MU grouping and pairing is enabled.
Uplink MU-MIMO for RedCap UEs	**REDCAP_UL_MU_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*RedCap*	Uplink MU grouping and pairing optimization for RedCap UEs can take effect only when uplink MU-MIMO for RedCap UEs is enabled.
RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	Uplink MU grouping and pairing optimization for RedCap UEs can be enabled only when RedCap Introduction and uplink MU grouping and pairing are enabled.
Uplink MU grouping and pairing	**UL_MU_GRP_PAIR_SW** option of the **NRDUCellUlMimo**.***UlMuMimoAlgoSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	

Function Name	Function Switch	Reference	Description
Uplink MU grouping and pairing optimization for RedCap UEs	**REDCAP_UL_MU_GRP_PAIR_OPT_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	Uplink MU grouping and pairing avoidance for RedCap UEs can take effect only when uplink MU grouping and pairing optimization for RedCap UEs is enabled.
Uplink basic preallocation	**UL_BASIC_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	Uplink preallocation period adjustment can take effect only when uplink basic preallocation or uplink smart preallocation is enabled.
Uplink smart preallocation	**UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	
Closed-loop power control for the PUSCH in dynamic scheduling mode	**PUSCH_DYN_SCH_CLOSED_LOOP_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoSwitch*** parameter	*Power Control*	Data-volume-based intelligent power control can take effect only when closed-loop power control for the PUSCH in dynamic scheduling mode is enabled.
Probability-based uplink MCS selection	**UL_PBMCSSEL_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter	*Turbo Uplink (Low-Frequency TDD)*	Adaptive outer-loop adjustment for small-packet services can take effect only when probability-based uplink MCS selection is enabled.
SR-based smart scheduling	**SR_SMART_SCH_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	*Uplink Scheduling*	MCS index optimization for SR-based scheduling can take effect only when SR-based smart scheduling is enabled.

Functions of control channel interference reduction require the following functions.

Function Name	Function Switch	Reference	Description
Cell slot configuration	**NRDUCell**.***SlotAssignment***	*Cell Management*	Smart PDCCH symbol allocation can be enabled only when all of the following conditions are met: ● The **NRDUCell**.***SlotAssignment*** parameter is set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**. ● The **UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter is selected. ● The **NRDUCellChnPwr**.***MaxPdcchAggPwrOffset*** parameter is set to a value greater than or equal to 30.
PDCCH symbol number adaptation	**UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter	*Channel Management*	
Maximum Aggregated PDCCH Power Offset	**NRDUCellChnPwr**.***MaxPdcchAggPwrOffset***	None	
LTE TDD and NR dynamic spectrum sharing	**LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None	If the **LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter is selected, smart PDCCH symbol allocation can be enabled only when any of the following conditions is met: ● **gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex*** – **gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex*** + 1 > 189 ● 106 ≤ **gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex*** – **gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex*** + 1 ≤ 189, and **NRDUCellCoreset**.***CommonCtrlResRbNum*** is set to **RB48**.
NR Reserved RB Start Index	**gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex***	None	
NR Reserved RB End Index	**gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex***	None	
Common Control Resource RB Number	**NRDUCellCoreset**.***CommonCtrlResRbNum***	None	

Function Name	Function Switch	Reference	Description
Super uplink	**SUPER_UL_TDM_SCH_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	Assume that any of the switches for super uplink, flexible super uplink, and inter-base-station super uplink is turned on. Smart PDCCH symbol allocation can be enabled only when the switch for PDCCH blind detection assignment optimization is turned on.
Flexible super uplink	**FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	
Inter-base-station super uplink	**INTER_GNB_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	None	
PDCCH blind detection assignment optimization	**PDCCH_BLIND_DET_ASSIGN_OPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	

Function Name	Function Switch	Reference	Description
Downlink cell bandwidth	**NRDUCell**.***DlBandwidth***	*Cell Management*	Assume that the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB48**. The **CCE_SYMBOL_ALLOC_OPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter can be selected only when any of the following conditions is met: ● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_40M**, **CELL_BW_50M**, **CELL_BW_60M**, or **CELL_BW_70M**, and the **LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter is deselected. ● The **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_40M**, **CELL_BW_50M**, **CELL_BW_60M**, **CELL_BW_70M**, **CELL_BW_80M**, **CELL_BW_90M**, or **CELL_BW_100M**, the **LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter is selected, and the result of (**gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex*** – **gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex*** + 1) is in the range of [106, 189].
LTE TDD and NR dynamic spectrum sharing	**LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None	
NR Reserved RB Start Index	**gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex***	None	
NR Reserved RB End Index	**gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex***	None	
Common Control Resource RB Number	**NRDUCellCoreset**.***CommonCtrlResRbNum***	None	
CSI reporting period adaptation	**CSI_REPORT_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch**.***CsiResoureAlgoSwitch*** parameter	*Channel Management*	BWP2 CSI period optimization can take effect only when CSI reporting period adaptation is enabled.

Function Name	Function Switch	Reference	Description
Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	• BWP2 CSI period optimization can be enabled only after power saving BWP is enabled. • BWP2 PUCCH frequency-domain position optimization can be enabled only after power saving BWP is enabled.

Functions of reference signal interference reduction require the following functions.

Function Name	Function Switch	Reference	Description
Time-domain adaptive filtering	**PUSCH_TIME_DOM_ADAPT_FILTER_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	IRC-based PDP detection can take effect only when time-domain adaptive filtering is enabled.

4.3.2.2 Mutually Exclusive Functions

Functions of traffic channel interference reduction have the following mutually exclusive functions.

Function Name	Function Switch	Reference	Description
MU non-alignment pairing	**MU_NONALIGNED_PAIR_SW** option of the **NRDUCellUlMimo**.***UlMuMimoAlgoSwitch*** parameter	None	Uplink MU grouping and pairing cannot be enabled when any of the following options is selected: • **MU_NONALIGNED_PAIR_SW** option of the **NRDUCellUlMimo**.***UlMuMimoAlgoSwitch*** parameter • **MU_DMRS_NONORTH_MULTIPLEX_SW** option of the **NRDUCellDmrs**.***UlDmrsSwitch*** parameter
MU DMRS non-orthogonal multiplexing	**MU_DMRS_NONORTH_MULTIPLEX_SW** option of the **NRDUCellDmrs**.***UlDmrsSwitch*** parameter	None	
High-speed cell	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	Uplink MU grouping and pairing cannot be enabled in high-speed cells.

Function Name	Function Switch	Reference	Description
Grant Free Resource Mode	**NRDUCellPusch**.***GrantFreeResourceMode***	None	Uplink MU grouping and pairing cannot be enabled when both of the following conditions are met: ● The **NRDUCellPusch**.***GrantFreeResourceMode*** parameter is set to **FREQUENCY_SHARING**. ● The **GRANT_FREE_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter is selected.
Grant-free	**GRANT_FREE_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter	None	
Uplink DMRS Type	**NRDUCellPusch**.***UlDmrsType*** set to **TYPE2**	*Uplink Scheduling*	DCI-based dynamic waveform switching cannot be enabled when uplink DMRS type 2 is used.
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	DCI-based dynamic waveform switching cannot be enabled in high-speed cells.

Functions of control channel interference reduction have the following mutually exclusive functions.

Function Name	Function Switch	Reference	Description
Scheduling in common PDCCH symbols for RedCap UEs	**REDCAP_COMM_PDCCH_SYM_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*RedCap*	Smart PDCCH symbol allocation and "scheduling in common PDCCH symbols for RedCap UEs" cannot be both enabled.
Bi-Carrier	**BI_CARRIER_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	None	Smart PDCCH symbol allocation and Bi-Carrier cannot be both enabled.

Function Name	Function Switch	Reference	Description
Super uplink	**SUPER_UL_TDM_SCH_SW**, **FLEX_SUPER_UPLINK_SW**, or **INTER_GNB_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	Smart PDCCH symbol allocation and super uplink cannot be both enabled.
Smart downlink repetition	**DL_SMART_REPETITION_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	None	Smart PDCCH symbol allocation and smart downlink repetition cannot be both enabled.
PDCCH multi-symbol resource pool	**MULTI_SYM_CORESET_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	*Channel Management*	Smart PDCCH symbol allocation and PDCCH multi-symbol resource pool cannot be both enabled.
Downlink semi-persistent scheduling	**DL_SPS_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	None	Smart PDCCH symbol allocation and downlink semi-persistent scheduling cannot be both enabled.
Average 3D EIRP control	**AVERAGE_3DEIRP_CONTROL_SW** option of the **NRDUCellSmartPwrLock**.***SmartPowerLockSw*** parameter	None	Smart PDCCH symbol allocation and average 3D EIRP control cannot be both enabled.
Fast CBF	**QUICK_CBF_SW** option of the **NRDUCellComp**.***CsCbfAlgoSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	Smart PDCCH symbol allocation and fast CBF cannot be both enabled.
Downlink single DCI	**DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	Smart PDCCH symbol allocation and downlink single DCI cannot be both enabled.

Function Name	Function Switch	Reference	Description
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**, and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** or **DPS_MODE**	*Cell Combination*	Smart PDCCH symbol allocation cannot be enabled when all the following conditions are met: ● The **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **HYPER_CELL_COMBINE_MODE**. ● The **NRDUCellMultiTrp**.***DataTransMode*** parameter is set to **BASIC_MODE** or **DPS_MODE**. ● The **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB48**.
Common Control Resource RB Number	**NRDUCellCoreset**.***CommonCtrlResRbNum***	None	
3D communication coverage mode	**NRDUCell**.***TridCommMode*** set to **TRID_COVERAGE_MODE**	None	Smart PDCCH symbol allocation cannot be enabled when both of the following conditions are met: ● The **NRDUCell**.***TridCommMode*** parameter is set to **TRID_COVERAGE_MODE**. ● The **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter is set to **RB48**.
Common Control Resource RB Number	**NRDUCellCoreset**.***CommonCtrlResRbNum***	None	

Functions of reference signal interference reduction have no mutually exclusive functions.

4.3.2.3 Function Impacts

Functions of traffic channel interference reduction have the following function impacts.

Function Name	Function Switch	Reference	Description
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When intra-base-station UL CoMP takes effect, enabling uplink MU grouping and pairing may decrease the value of the **N.IntragNB.UL.CoMP.SchUser.Sum** counter.

Function Name	Function Switch	Reference	Description
Interference randomization under weak uplink coverage	**UL_WEAK_COV_INTRF_RAND_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	After uplink MU grouping and pairing takes effect, the proportion of weak-coverage UEs for which frequency selective scheduling is performed may decrease.
Avoidance of RedCap uplink scheduling	**NRDUCellRedCap**.***UlAvoidPol***	*Stable Video Surveillance Experience*	If the avoidance of RedCap uplink scheduling function is enabled (that is, the **NRDUCellRedCap**.***UlAvoidPol*** parameter is set to **LEVEL1** or **LEVEL2**) and uplink MU grouping and pairing optimization for RedCap UEs is enabled, the number of uplink RBs that can be used for avoidance in the activated BWP1s for RedCap UEs may be limited. This function may not produce gains for RedCap UEs.
SR period adaptation	**SR_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch**.***SrResoureAlgoSwitch*** parameter	*Channel Management*	When the load is heavy in the uplink and there are a large number of UEs, the proportion of CCE resources used for data scheduling decreases due to an increase in the number of SRs. SR period adaptation is recommended when packet-length-based scheduling optimization is enabled.
PUCCH MRC/IRC adaptation	**PUCCH_MRC_IRC_SW** option of the **NRDUCellPucch**.***PucchReceiveEnhSwitch*** parameter	*Channel Management*	When the load is heavy in the uplink, the uplink interference increases. As a result, the PUCCH demodulation performance deteriorates. PUCCH MRC/IRC adaptation is recommended when packet-length-based scheduling optimization is enabled.

Function Name	Function Switch	Reference	Description
CSI reporting period adaptation	**CSI_REPORT_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch**.***CsiResoureAlgoSwitch*** parameter	*Channel Management*	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, it is recommended that CSI reporting period adaptation be enabled to increase the proportion of UEs with periodic CSIs and reduce PDCCH resource consumption.
Dynamic PDCCH SINR adjustment based on PUSCH DTX	**PUSCH_DTX_AGG_LVL_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	*Channel Management*	When the load is heavy in the uplink, the uplink interference increases and the number of PUSCH DTXs increases. As a result, CCE aggregation level selection is inaccurate. When packet-length-based scheduling optimization is enabled, it is recommended that dynamic PDCCH SINR adjustment based on PUSCH DTX be enabled. This allows the estimated PDCCH SINR to be dynamically adjusted based on the PUSCH DTX information so that a more appropriate CCE aggregation level can be selected.

Function Name	Function Switch	Reference	Description
Initial CCE aggregation level selection optimization	**AGG_LVL_INIT_SELECT_OPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	*Channel Management*	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, it is recommended that initial CCE aggregation level selection optimization be enabled. This reduces the CCE aggregation level, increases the UE capacity that can be accommodated by PDCCH resources, and reduces the CCE allocation failure rate.
PDCCH Aggregation Level Adaptation Policy	**NRDUCellPdcch**.***PdcchAggLvlAdaptPol*** set to **SEPARATE_ADAPT**	*Channel Management*	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, it is recommended that the PDCCH aggregation level adaptation policy be used. This allows the aggregation level selected for the UE-specific PDCCH to be more accurate and reduces the CCE allocation failure rate, when the channel quality changes rapidly.

Function Name	Function Switch	Reference	Description
PDCCH uplink-to-downlink CCE ratio adaptation	**UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	*Channel Management*	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, it is recommended that PDCCH uplink-to-downlink CCE ratio adaptation be enabled. This increases the proportion of PDCCH resources used for uplink scheduling but may decrease the proportion of PDCCH resources used for downlink scheduling.
PDCCH symbol number adaptation	**UE_PDCCH_SYM_NUM_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoExtSwitch*** parameter	*Channel Management*	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. When packet-length-based scheduling optimization is enabled, it is recommended that PDCCH symbol number adaptation be enabled to increase the number of PDCCH resources. In addition, it is recommended that the **NRDUCellPdcch**.***HeavyLoadCceUsageThld*** parameter be set to **50**.

Function Name	Function Switch	Reference	Description
Power Saving BWP CCE Adjustment Policy	**NRDuCellPdcch**.***PwrSaveBwpCceAdjustPolicy*** set to **PWR_SAVING_BWP_NOT_CONSIDERED**	*UE Power Saving*	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. With packet-length-based scheduling optimization enabled, it is recommended that the **NRDuCellPdcch**.***PwrSaveBwpCceAdjustPolicy*** parameter be set to **PWR_SAVING_BWP_NOT_CONSIDERED** if the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter is selected. This prevents UEs involved in power saving BWP from participating in CCE ratio adjustment.
Adaptive awareness scheduling	**ADAPT_AWARE_SCH_SW** option of the **NRDUCellUlSch**.***UlSchAlgoSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	Packet-length-based scheduling optimization does not take effect when it is enabled together with adaptive awareness scheduling.
Uplink precise MCS optimization	**UL_PRECISE_MCS_OPT_SW** option of the **NRDUCellAiAlgo**.***AiAmcAlgoSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	Uplink precise MCS optimization preferentially takes effect when it is enabled together with "adaptive outer-loop adjustment for small-packet services".

Functions of control channel interference reduction have the following function impacts.

Function Name	Function Switch	Reference	Description
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When DL CoMP takes effect, enabling smart PDCCH symbol allocation may decrease the uplink CCE allocation success rate of UEs and the average uplink UE throughput.

Function Name	Function Switch	Reference	Description
RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	When smart PDCCH symbol allocation is enabled in scenarios where RedCap Introduction takes effect, the CCE allocation success rate may increase for RedCap UEs at the cell center or at a medium distance from the cell center, but it may decrease for RedCap UEs at the cell edge. As a result, the average uplink and downlink UE throughputs may change.
RF channel intelligent shutdown	**RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	When RF channel intelligent shutdown takes effect, enabling smart PDCCH symbol allocation may result in a smaller increase in the uplink CCE allocation success rate for UEs and therefore may deliver lower gains in the average uplink UE throughput.
Channel adaptation optimization	**CHANNEL_ADAPT_OPT_SW** option of the **NRDUCellIntrfIdent**.***ChnMeasCtrlSw*** parameter	*LTE and NR Zero Bufferzone*	When channel adaptation optimization in "LTE and NR Zero Bufferzone" takes effect, enabling smart PDCCH symbol allocation may result in a smaller increase in the uplink CCE allocation success rate and therefore may deliver lower gains in the average uplink UE throughput.
NR inter-carrier dynamic power sharing	**NR_DYN_POWER_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***DynPowerSharingSwitch*** parameter	*NR Inter-Carrier Dynamic Power Allocation*	When enabled together with smart PDCCH symbol allocation, NR inter-carrier dynamic power sharing may take effect in a lower proportion of cases, possibly providing lower downlink gains.

Function Name	Function Switch	Reference	Description
Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*CA Experience Improvement*	When inter-gNodeB CA takes effect for a UE with smart PDCCH symbol allocation enabled for SCells but not the PCell, the average downlink throughput of the inter-gNodeB CA UE may decrease.
Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	● Enabling smart PDCCH symbol allocation decreases the uplink CCE allocation success rate in a cell when all the following conditions are met in the cell: (1) Power saving BWP takes effect. (2) The duration in which UEs working on BWP2s accounts for a large proportion. (3) The **NRDUCellUePwrSaving**.***Bwp2MaxRbNum*** parameter is set to **RB64** for BWP2s. ● When smart PDCCH symbol allocation is enabled together with power saving BWP, the CCE allocation success rate of BWP2 UEs increases. As a result, more UEs may work on BWP2s, and the average uplink UE throughput does not increase or even decreases. To resolve this issue, you can turn off the BWP2 switch or set the **NRDUCellUePwrSaving**.***Bwp2SchFailRateThld*** parameter to a smaller value (for example, **1** when the downlink CCE allocation success rate is lower than 80%).

Functions of reference signal interference reduction have no function impacts.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- For adaptive outer-loop adjustment for small-packet services, uplink MU grouping and pairing, mixed pairing of UEs using different waveforms, smart PDCCH symbol allocation, and IRC-based PDP detection:

  3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- For other functions:
  - 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
  - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- For adaptive outer-loop adjustment for small-packet services, uplink MU grouping and pairing, mixed pairing of UEs using different waveforms, and smart PDCCH symbol allocation: All NR TDD-capable low-frequency RF modules with at least 32T32R can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- For IRC-based PDP detection: All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- For other functions:
  - Macro: All NR TDD-capable low-frequency RF modules with at least 32T32R can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
  - LampSite: The RF module requirements are the same as those of distributed massive MIMO. For details about the RF module requirements of distributed massive MIMO, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

4.3.4 Cells

Uplink Boosting has the following cell requirements:

- The **NRDUCell**.***FrequencyBand*** parameter must be set to **N38**, **N40**, **N41**, **N77**, **N78**, or **N79**.
- When the **NRDUCell**.***LampSiteCellFlag*** parameter is set to **YES**, the **NRDUCellTrp**.***TrpType*** parameter must be set to **MASTER_DM_MIMO** or **MASTER_DM_MIMO_LIGHT**.
- It is recommended that the **NRDUCell**.***NrDuCellNetworkingMode*** parameter be set to **NORMAL_CELL**.
- The **NRDUCell**.***DuplexMode*** parameter cannot be set to **CELL_SDL**.

In addition to the overall cell requirements of Uplink Boosting, a cell must meet the following conditions if smart PDCCH symbol allocation is required:

- The **NRDUCell**.***DlBandwidth*** parameter must be set to **CELL_BW_80M**, **CELL_BW_90M**, or **CELL_BW_100M**.
- The **NRDUCell**.***DlBandwidth*** parameter must be set to **CELL_BW_40M**, **CELL_BW_50M**, **CELL_BW_60M**, or **CELL_BW_70M**, and the **NRDUCellCoreset**.***CommonCtrlResRbNum*** parameter must be set to **RB48**.

In addition to the overall cell requirements of Uplink Boosting, the **NRDUCell**.***UlBandwidth*** parameter must be set to a value greater than or equal to 40 MHz if uplink MU grouping and pairing is required.

4.3.5 Others

DCI-based dynamic waveform switching requires support from UEs. Whether a UE supports this function in a band depends on the number of carriers configured in the band and the UE's support for the *dynamicWaveformSwitch-r18* and *dynamicWaveformSwitchIntraCA-r18* fields.

- If only one carrier is configured in the band, the UE supports this function as long as it supports the *dynamicWaveformSwitch-r18* field.
- If multiple carriers are configured in the band, the UE supports this function only when the value of the *dynamicWaveformSwitchIntraCA-r18* field in the UE capability is greater than or equal to the number of carriers configured in the band.

For details about the *dynamicWaveformSwitch-r18* and *dynamicWaveformSwitchIntraCA-r18* fields, see section 4.2.7.2 "BandNR parameters" in 3GPP TS 38.306 V18.2.0.

NOTE

If a UE has carriers configured in different bands, whether DCI-based dynamic waveform switching can take effect for each band is separately determined based on the preceding conditions. For example, if a UE meets the conditions for this function to take effect in band A but not in band B, this function can take effect in band A but not in band B.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 describes the parameters used for activating Uplink Boosting.

Table 4-1 Parameters used for activating Uplink Boosting

Parameter Name	Parameter ID	Option	Setting Notes
MIMO Feature Switch	**NRDUCellFeatureSw**.***MimoFeatureSwitch***	UL_LOW_NOISE_SW	Select this option when Uplink Boosting is required.

Table 4-2 describes the parameters used for activating functions of traffic channel interference reduction.

Table 4-2 Parameters used for activating functions of traffic channel interference reduction

Parameter Name	Parameter ID	Option	Setting Notes
UL MU-MIMO Algorithm Switch	**NRDUCellUlMimo**.***UlMuMimoAlgoSwitch***	UL_MU_GRP_PAIR_SW	Select this option if uplink MU grouping and pairing is required.
Flexible Pairing RB Ratio Threshold	**NRDUCellUlMimo**.***FlexPairRbRatioThld***	None	Set this parameter based on the network plan if uplink MU grouping and pairing is required.
MU-MIMO Grp Pairing Norm SU Min RB Ratio	**NRDUCellUlMimo**.***MuGrpPairNormSuMinRbRatio***	None	Set this parameter based on the network plan if uplink MU grouping and pairing is required.
RedCap Schedule Algorithm Switch	**NRDUCellRedCap**.***RedCapSchAlgoSwitch***	REDCAP_UL_MU_GRP_PAIR_OPT_SW	Select this option if uplink MU grouping and pairing is required for RedCap UEs.
RedCap Schedule Algorithm Switch	**NRDUCellRedCap**.***RedCapSchAlgoSwitch***	REDCAP_UL_MU_GRP_PAIR_AVOID_SW	Select this option if uplink MU grouping and pairing is required for RedCap UEs.
UL MU-MIMO Algorithm Switch	**NRDUCellUlMimo**.***UlMuMimoAlgoSwitch***	DIFF_WAVEFORM_PAIR_SW	Select this option if UEs using different waveforms for scheduling need to participate in uplink MU grouping and pairing.

Parameter Name	Parameter ID	Option	Setting Notes
UL MU-MIMO Algorithm Switch	**NRDUCellUlMimo**.***UlMuMimoAlgoSwitch***	UL_CORR_ACCELERATION_SW	Select this option if optimized uplink correlation-based pairing is required.
PUSCH Algorithm Extension Switch	**NRDUCellPusch**.***PuschAlgoExtSwitch***	PUSCH_RES_ADAPT_ALLOC_SW	Select this option if adaptive PUSCH resource allocation is required.
Uplink Preallocation Switch	**NRDUCellPusch**.***UlPreallocationSwitch***	UL_PREALLOCATION_PERIOD_ADJ_SW	Select this option if uplink preallocation period adjustment is required.
UL Power Control Algorithm Extension Switch	**NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch***	DATA_VOL_INTEL_PWR_CTRL_SW	Select this option if data-volume-based intelligent power control is required.
UL Power Control Algorithm Extension Switch	**NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch***	DATA_VOL_INTEL_PWR_CTRL_OPT_SW	Select this option if data-volume-based intelligent power control is required.
Small Packet UE PUSCH Power Offset	**NRDUCellUlPcConfig**.***SmallPktUePuschPwrOffset***	None	Set this parameter based on the network plan if data-volume-based intelligent power control is required.
Small Packet UE SRS SINR Thld	**NRDUCellUlPcConfig**.***SmallPktUeSrsSinrThld***	None	Set this parameter based on the network plan if data-volume-based intelligent power control is required.
PUSCH Performance Switch	**NRDUCellPusch**.***PuschPerformanceSwitch***	PKT_LEN_BASED_SCH_OPT_SW	Select this option if packet-length-based scheduling optimization is required. This function is not recommended in scenarios where CCE resources are insufficient or there are a large number of small-packet UEs, since it may decrease the average uplink UE throughput.
Uplink AMC Algorithm Switch	**NRDUCellUlAmc**.***UlAmcAlgoSw***	UL_CELL_OLLA_SW	Select this option if uplink cell-level OLLA is required.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Performance Switch	**NRDUCellUlAmc**.***UlAmcPerformanceSw***	SMALL_PKT_OL_ADAPT_ADJ_SW	Select this option if adaptive outer-loop adjustment for small-packet services is required.
Outer-Loop Adaptive Adjustment SINR Thld	**NRDUCellUlAmc**.***OlAdaptAdjSinrThld***	None	Set this parameter based on the network plan if adaptive outer-loop adjustment for small-packet services is required.
Uplink AMC Performance Switch	**NRDUCellUlAmc**.***UlAmcPerformanceSw***	SR_BASED_SCH_MCS_OPT_SW	Select this option if MCS index optimization for SR-based scheduling is required.
UL Coverage Algorithm Switch	**NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch***	DYNAMIC_WAVEFORM_SWITCHING_SW	Select this option if DCI-based dynamic waveform switching is required.
SINR Threshold for Waveform Selection	**NRDUCellPusch**.***SinrThldforWaveformSel***	None	Set this parameter based on the network plan if DCI-based dynamic waveform switching is required.

Table 4-3 describes the parameters used for activating functions of control channel interference reduction.

Table 4-3 Parameters used for activating functions of control channel interference reduction

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Enhancement Switch	**NRDUCellPdcch**.***PdcchAlgoEnhSwitch***	PDCCH_SYM_SMART_ALLOC_SW	Select this option if smart PDCCH symbol allocation is required.
PUCCH Algorithm Switch	**NRDUCellPucch**.***PucchAlgoSwitch***	CCE_SYMBOL_ALLOC_OPT_SW	Select this option if smart PDCCH symbol allocation is required in a cell with a bandwidth of 40 MHz to 70 MHz.
PUCCH Algorithm Switch	**NRDUCellPucch**.***PucchAlgoSwitch***	PDCCH_BLIND_DET_ASSIGN_OPT_SW	Select this option if PDCCH blind detection assignment optimization is required.

Parameter Name	Parameter ID	Option	Setting Notes
Blacklist Control Extension Switch	**gNBUeCompatSw**.***BlacklistCtrlExtSwitch***	PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF	Select this option if there are UEs incompatible with smart PDCCH symbol allocation on the network.
CSI Resource Algo Switch	**NRDUCellPucch**.***CsiResoureAlgoSwitch***	BWP2_CSI_PERIOD_OPT_SWITCH	Select this option if BWP2 CSI period optimization is required.
PUCCH Algorithm Switch	**NRDUCellPucch**.***PucchAlgoSwitch***	BWP2_PUCCH_POS_OPT_SW	Select this option if BWP2 PUCCH frequency-domain position optimization is required.
PUCCH Algorithm Extension Switch	**NRDUCellPucch**.***PucchAlgoExtSwitch***	F1_ACK_CODE_CHN_INTRF_OPT_SW	Select this option if PUCCH F1 ACK code channel interference avoidance is required.

Table 4-4 describes the parameters used for activating functions of reference signal interference reduction.

Table 4-4 Parameters used for activating functions of reference signal interference reduction

Parameter Name	Parameter ID	Option	Setting Notes
UL DMRS Switch	**NRDUCellDmrs**.***UlDmrsSwitch***	IRC_BASED_PDP_DETECT_SW	Select this option if IRC-based PDP detection is required.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Enabling Uplink Boosting

```
//Turning on the switch for Uplink Boosting
MOD NRDUCELLFEATURESW: NrDuCellId=0, MimoFeatureSwitch=UL_LOW_NOISE_SW-1;
```

Enabling functions of traffic channel interference reduction

```
//Enabling uplink MU grouping and pairing
MOD NRDUCELLULMIMO: NrDuCellId=0, UlMuMimoAlgoSwitch=UL_MU_GRP_PAIR_SW-1,
FlexPairRbRatioThld=100, MuGrpPairNormSuMinRbRatio=20;
//(Optional) Enabling uplink MU grouping and pairing for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0,
RedCapSchAlgoSwitch=REDCAP_UL_MU_GRP_PAIR_OPT_SW-1&REDCAP_UL_MU_GRP_PAIR_AVOID_SW-1;
//Enabling mixed pairing of UEs using different waveforms
MOD NRDUCELLULMIMO: NrDuCellId=0, UlMuMimoAlgoSwitch=DIFF_WAVEFORM_PAIR_SW-1;
//Enabling optimized uplink correlation-based pairing
MOD NRDUCELLULMIMO: NrDuCellId=0, UlMuMimoAlgoSwitch=UL_CORR_ACCELERATION_SW-1;
//Enabling adaptive PUSCH resource allocation
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=PUSCH_RES_ADAPT_ALLOC_SW-1;
//Enabling uplink preallocation period adjustment
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPreallocationSwitch=UL_PREALLOCATION_PERIOD_ADJ_SW-1;
//Enabling data-volume-based intelligent power control
MOD NRDUCELLULPCCONFIG: NrDuCellId=0,
UlPwrCtrlAlgoExtSwitch=DATA_VOL_INTEL_PWR_CTRL_SW-1&DATA_VOL_INTEL_PWR_CTRL_OPT_SW-1,
SmallPktUePuschPwrOffset=3, SmallPktUeSrsSinrThld=0;
//(Optional) Enabling packet-length-based scheduling optimization. However, this function is not
recommended in scenarios where CCE resources are insufficient or there are a large number of small-packet
UEs, since it may decrease the average uplink UE throughput.
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=PKT_LEN_BASED_SCH_OPT_SW-1;
//Enabling uplink cell-level OLLA
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcAlgoSw=UL_CELL_OLLA_SW-1;
//Enabling adaptive outer-loop adjustment for small-packet services
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcPerformanceSw=SMALL_PKT_OL_ADAPT_ADJ_SW-1,
OlAdaptAdjSinrThld=8;
//Enabling MCS index optimization for SR-based scheduling
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcPerformanceSw=SR_BASED_SCH_MCS_OPT_SW-1;
//Enabling DCI-based dynamic waveform switching
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,
UlCoverageAlgoSwitch=DYNAMIC_WAVEFORM_SWITCHING_SW-1;
MOD NRDUCELLPUSCH: NrDuCellId=0, SinrThldforWaveformSel=32;
```

Enabling functions of control channel interference reduction

```
//Enabling smart PDCCH symbol allocation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_SYM_SMART_ALLOC_SW-1;
//(Optional) Enabling smart PDCCH symbol allocation for a cell with a bandwidth of 40 MHz to 70 MHz
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=CCE_SYMBOL_ALLOC_OPT_SW-1;
//(Optional) Enabling smart PDCCH symbol allocation when the assignment of blind detections across
different aggregation levels is uneven for the PDCCH
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_BLIND_DET_ASSIGN_OPT_SW-1;
//Enabling blacklist control for dynamic PDCCH symbol allocation when some UEs on the network do not
support smart PDCCH symbol allocation (with ec7fee769ff7e0fff0 used as an example of the value of
UeFeatureValueContent)
ADD GNBUEINFO: UeInfoIndex=1, UeInfoType=UE_FEATUREVALUE,
UeFeatureValueContent="ec7fee769ff7e0fff0",
UeFeatureValueMask="FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF", UeFeatureValueVersion=VERSION1;
ADD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF-1;
//(Optional) Enabling BWP2 CSI period optimization after power saving BWP is enabled for the cell
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiResoureAlgoSwitch=BWP2_CSI_PERIOD_OPT_SWITCH-1;
//(Optional) Enabling BWP2 PUCCH frequency-domain position optimization after power saving BWP is
enabled for the cell
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=BWP2_PUCCH_POS_OPT_SW-1;
//Enabling PUCCH F1 ACK code channel interference avoidance
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=F1_ACK_CODE_CHN_INTRF_OPT_SW-1;
```

Enabling functions of reference signal interference reduction

```
//Enabling IRC-based PDP detection
MOD NRDUCELLDMRS: NrDuCellId=0, UlDmrsSwitch=IRC_BASED_PDP_DETECT_SW-1;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Disabling functions of reference signal interference reduction

```
//Disabling IRC-based PDP detection
MOD NRDUCELLDMRS: NrDuCellId=0, UlDmrsSwitch=IRC_BASED_PDP_DETECT_SW-0;
```

Disabling functions of control channel interference reduction

```
//Disabling PUCCH F1 ACK code channel interference avoidance
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=F1_ACK_CODE_CHN_INTRF_OPT_SW-0;
//Disabling BWP2 PUCCH frequency-domain position optimization
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoSwitch=BWP2_PUCCH_POS_OPT_SW-0;
//Disabling BWP2 CSI period optimization
MOD NRDUCELLPUCCH: NrDuCellId=0, CsiResoureAlgoSwitch=BWP2_CSI_PERIOD_OPT_SWITCH-0;
//Disabling blacklist control for dynamic PDCCH symbol allocation
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=PDCCH_SYM_DYNAMIC_ALLOC_SW_OFF-0;
//(Optional) Disabling smart PDCCH symbol allocation for the cell with a bandwidth of 40 MHz to 70 MHz
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=CCE_SYMBOL_ALLOC_OPT_SW-0;
//Disabling PDCCH blind detection assignment optimization
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoSwitch=PDCCH_BLIND_DET_ASSIGN_OPT_SW-0;
//Disabling smart PDCCH symbol allocation
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_SYM_SMART_ALLOC_SW-0;
```

Disabling functions of traffic channel interference reduction

```
//Disabling DCI-based dynamic waveform switching
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,
UlCoverageAlgoSwitch=DYNAMIC_WAVEFORM_SWITCHING_SW-0;
//Disabling MCS index optimization for SR-based scheduling
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcPerformanceSw=SR_BASED_SCH_MCS_OPT_SW-0;
//Disabling adaptive outer-loop adjustment for small-packet services
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcPerformanceSw=SMALL_PKT_OL_ADAPT_ADJ_SW-0;
//Disabling uplink cell-level OLLA
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcAlgoSw=UL_CELL_OLLA_SW-0;
//Disabling packet-length-based scheduling optimization
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschPerformanceSwitch=PKT_LEN_BASED_SCH_OPT_SW-0;
//Disabling data-volume-based intelligent power control
MOD NRDUCELLULPCCONFIG: NrDuCellId=0,
UlPwrCtrlAlgoExtSwitch=DATA_VOL_INTEL_PWR_CTRL_SW-0&DATA_VOL_INTEL_PWR_CTRL_OPT_SW-0;
//Disabling uplink preallocation period adjustment
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPreallocationSwitch=UL_PREALLOCATION_PERIOD_ADJ_SW-0;
//Disabling adaptive PUSCH resource allocation
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=PUSCH_RES_ADAPT_ALLOC_SW-0;
//Disabling optimized uplink correlation-based pairing
MOD NRDUCELLULMIMO: NrDuCellId=0, UlMuMimoAlgoSwitch=UL_CORR_ACCELERATION_SW-0;
//Disabling mixed pairing of UEs using different waveforms
MOD NRDUCELLULMIMO: NrDuCellId=0, UlMuMimoAlgoSwitch=DIFF_WAVEFORM_PAIR_SW-0;
//Disabling uplink MU grouping and pairing optimization for RedCap UEs
MOD NRDUCELLREDCAP: NrDuCellId=0,
RedCapSchAlgoSwitch=REDCAP_UL_MU_GRP_PAIR_OPT_SW-0&REDCAP_UL_MU_GRP_PAIR_AVOID_SW-0;
//Disabling uplink MU grouping and pairing
MOD NRDUCELLULMIMO: NrDuCellId=0, UlMuMimoAlgoSwitch=UL_MU_GRP_PAIR_SW-0;
```

Disabling Uplink Boosting

```
//Turning off the switch for Uplink Boosting
MOD NRDUCELLFEATURESW: NrDuCellId=0, MimoFeatureSwitch=UL_LOW_NOISE_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Functions of traffic channel interference reduction have taken effect in a cell if the average uplink interference or the number of RRC reconfiguration messages decreases in the cell after these functions are enabled.

Functions of control channel interference reduction have taken effect in a cell if the uplink CCE allocation success rate or average PUCCH SINR increases in the cell after these functions are enabled.

Functions of reference signal interference reduction have taken effect in a cell if the average uplink MCS index increases in the cell after these functions are enabled.

NOTE

Average PUCCH SINR = [**N.UL.SINR.PUCCH.Index0** x (–5) + **N.UL.SINR.PUCCH.Index1** x (–3) + **N.UL.SINR.PUCCH.Index2** + **N.UL.SINR.PUCCH.Index3** x 5 + **N.UL.SINR.PUCCH.Index4** x 9 + **N.UL.SINR.PUCCH.Index5** x 13 + **N.UL.SINR.PUCCH.Index6** x 17 + **N.UL.SINR.PUCCH.Index7** x 19]/(**N.UL.SINR.PUCCH.Index0** + **N.UL.SINR.PUCCH.Index1** + **N.UL.SINR.PUCCH.Index2** + **N.UL.SINR.PUCCH.Index3** + **N.UL.SINR.PUCCH.Index4** + **N.UL.SINR.PUCCH.Index5** + **N.UL.SINR.PUCCH.Index6** + **N.UL.SINR.PUCCH.Index7**)

4.4.3 Network Monitoring

After Uplink Boosting is enabled, use the performance indicators described in **4.2.1 Benefits** and **4.2.2 Impacts** to monitor cell performance.

5 Uplink Boosting 2.0

5.1 Principles

With the rapid rise in both the number of 5G UEs and traffic volume, the uplink load in NR cells rises, leading to a decline in uplink user experience in medium- and heavy-load scenarios. Uplink Boosting 2.0 is designed to address this issue. **This feature enhances the uplink performance of TDD massive MIMO cells through multi-cell power control for interference reduction, time-frequency scheduling for interference avoidance, and multi-domain demodulation for interference resistance, thereby increasing the average uplink UE throughput in medium- and heavy-load scenarios.** This feature is controlled by the **UL_LOW_NOISE_PHASE2_SW** option of the **NRDUCellFeatureSw**.***MimoFeatureSwitch*** parameter.

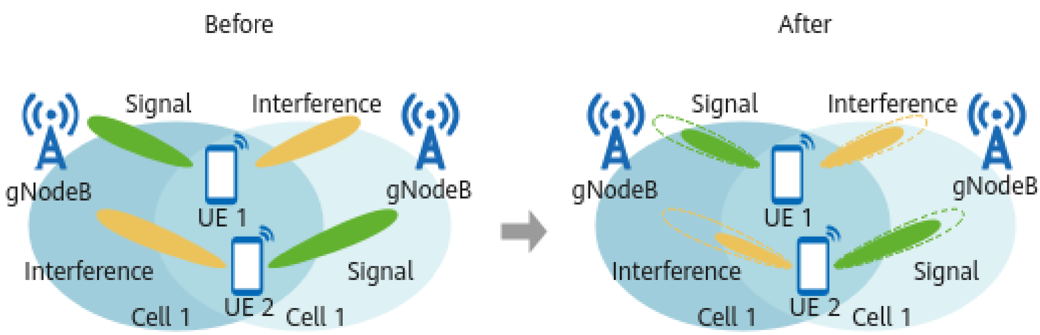
5.1.1 Multi-Cell Power Control for Interference Reduction

5.1.1.1 PUSCH Coordinated Power Control

Uplink interference is severe between medium- or heavy-load cells providing contiguous coverage. Severe interference degrades SINRs, leading to higher PUSCH transmit power according to the SINR-based PUSCH closed-loop power control algorithm. This increased power further exacerbates interference. To address this issue, PUSCH coordinated power control has been introduced.

This function allows the gNodeB to adjust the PUSCH transmit power of UEs based on UE performance in the serving cell and interference to intra-frequency neighboring cells. Figure 5-1 shows how this function works.

Figure 5-1 PUSCH coordinated power control



The gNodeB selects UEs that require transmit power adjustment in the serving cell and uses transmit power control (TPC) commands to adjust their PUSCH transmit power, thereby reducing interference between cells providing contiguous coverage. This operation is performed when both of the following conditions are met:

- Interference over thermal (IoT, which indicates the increase in the uplink interference and noise) in the cell ≥ **NRDUCellUlPcConfig**.***PuschJointPcCellIotThld***
- PUSCH PRB usage of the cell ≥ **NRDUCellUlPcConfig**.***PuschCoordPcPrbThld***

A target UE must meet all the following conditions:

- Measured SSB RSRP of the UE in the serving cell > **NRCellMeasConfig**.***UlCpcUeSsbRsrpThld***
- Measured SRS RSRP of the UE in the serving cell > **NRDUCellUlPcConfig**.***UlCpcUeSrsRsrpThld***
- Uplink interference of the UE to neighboring cells > **NRDUCellUlPcConfig**.***UlCpcUeIntrfRsrpThld***

UEs running special services are not selected as target UEs. Examples of special UEs include UEs requiring connectivity with low latency and high reliability, Cloud VR UEs, live streaming UEs, VoNR UEs, ViNR UEs, and Non-Slot UEs.

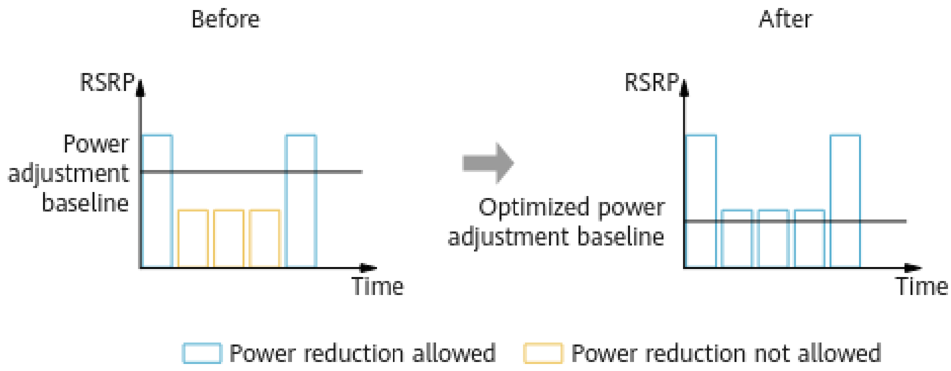
PUSCH coordinated power control is controlled by the **PUSCH_COORD_PWR_CTRL_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch*** parameter. It is recommended that this function be enabled in medium- and heavy-load scenarios.

5.1.1.2 Link-Performance-based Power Control

Under medium- or heavy-load conditions, cells that provide contiguous coverage face severe uplink inter-cell interference. If UEs at the cell center transmit signals at excessively high power levels, they will interfere with UEs in neighboring cells and affect their uplink user experience. Link-performance-based power control has been introduced to address this issue.

This function enables the gNodeB to consider UEs' link performance and power headroom during power adjustment. Specifically, the gNodeB reduces the PUSCH transmit power of UEs based on their uplink RSRP to mitigate inter-UE interference. Figure 5-2 shows how this function works.

Figure 5-2 Link-performance-based power control



This function is not applicable to special UEs such as UEs requiring connectivity with low latency and high reliability, Cloud VR UEs, live streaming UEs, VoNR UEs (with a 5QI of 1), ViNR UEs (with a 5QI of 2), Non-Slot UEs, and UEs requiring stable uplink service experience (with specified QCIs).

This function is controlled by the **INTRF_SC_LINK_PERF_PC_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch*** parameter. The **NRDUCellUlPcConfig**.***LinkPerfPcRsrpThldOffset*** parameter determines how much the PUSCH transmit power is reduced.

5.1.2 Time-Frequency Scheduling for Interference Avoidance

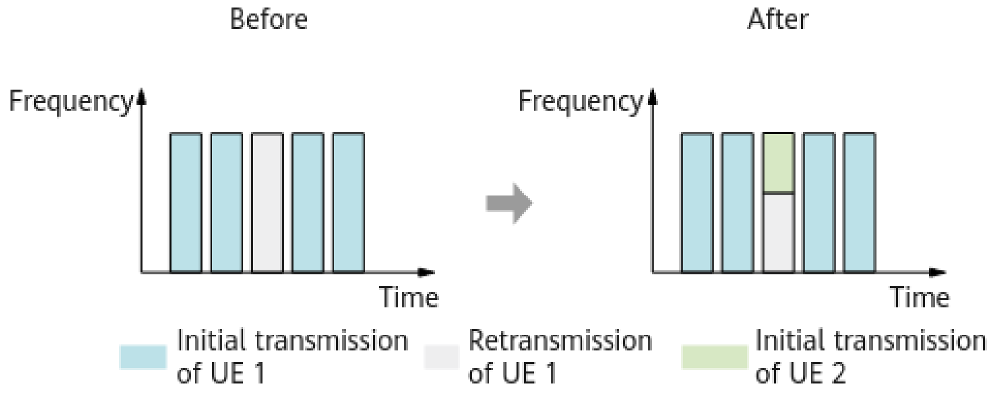
5.1.2.1 Precise RB Control for Uplink Retransmissions

Given the same MCS, if uplink retransmissions use fewer RBs than initial transmissions while maintaining a margin for HARQ combining gains, fewer resources are consumed for uplink retransmissions in multi-UE scenarios, freeing up more RBs for UEs performing initial transmissions on the PUSCH. This increases the average uplink UE throughput. Precise RB control for uplink retransmissions has been introduced to achieve this.

The precise RB control for uplink retransmissions function reduces the number of RBs for the first retransmission of UEs and reallocates the freed-up RBs to UEs performing initial transmissions on the PUSCH. This increases the average uplink UE throughput in medium- and heavy-load scenarios.

Figure 5-3 shows how this function works.

Figure 5-3 Precise RB control for uplink retransmissions



This function is controlled by the **UL_RETRANS_PREC_RB_CTRL_SW** option of the **NRDUCellPusch.*PuschAlgoExtSwitch*** parameter. The **NRDUCellUlSch.*UlFirstRetransMinRbPct*** parameter determines the percentage by which the number of retransmission RBs is reduced. This parameter must be set to a value other than **100**, and the recommended value is **20**.

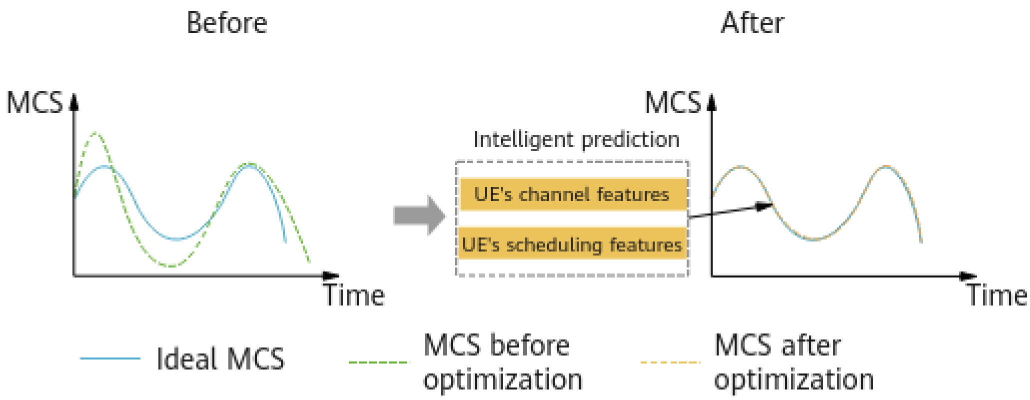
5.1.2.2 Uplink Precise MCS Optimization

On the live network, the transmission quality over channels fluctuates rapidly, preventing the selection of optimal MCSs for UEs. To address this challenge, uplink precise MCS optimization has been introduced.

This function enables the gNodeB to use a predictive model to determine an intelligent-optimization-based uplink MCS for a UE, accounting for the UE's channel and scheduling features. If the MCS prediction error is less than or equal to the predefined upper limit for uplink precise MCS optimization, this predicted MCS is used for uplink scheduling to increase the average uplink UE throughput.

Figure 5-4 shows how this function works.

Figure 5-4 Uplink precise MCS optimization



This function is controlled by the **UL_PRECISE_MCS_OPT_SW** option of the **NRDUCellAiAlgo.*AiAmcAlgoSwitch*** parameter. The **NRDUCellAiAlgo.*UlMcsPredictErrUpLimit*** parameter specifies the upper limit to the MCS prediction error for uplink precise MCS optimization. You can run the **DSP NRDUCELLAISCHMODEL** command to query model usage.

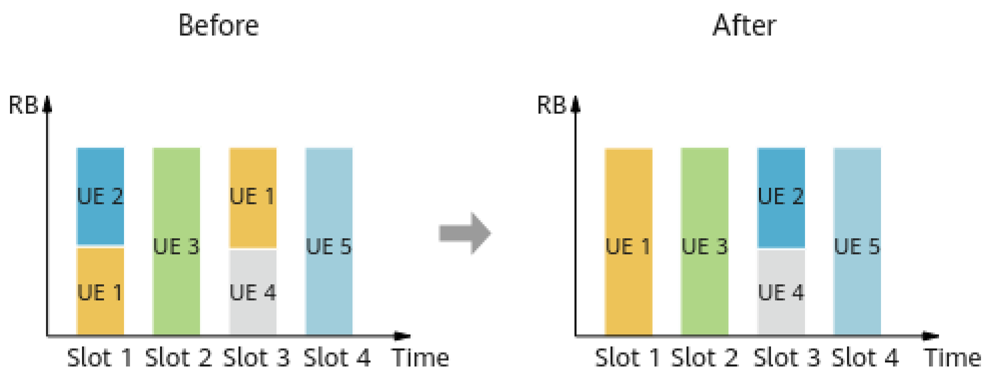
5.1.2.3 Adaptive Awareness Scheduling

Under medium- or heavy-load conditions on a live network, UEs' traffic models differ greatly and the transmission quality over channels fluctuates significantly. To shorten data transmission delays in such scenarios, it is essential to adjust UEs' uplink scheduling priorities adaptively based on specific traffic models. This challenge can be addressed through adaptive awareness scheduling.

This function enables the gNodeB to flexibly adjust the uplink scheduling priorities of UEs based on the RBs that they occupy and their scheduling delays. The purpose is to achieve consecutive scheduling for UEs whenever possible and minimize the data transmission delay, thereby increasing the average uplink UE throughput.

Figure 5-5 shows how this function works. With this function, UE 1's data that used to be scheduled in slot 3 is now scheduled in slot 1. As a result, UE 1's data transmission delay is reduced.

Figure 5-5 Adaptive awareness scheduling



This function is controlled by the **ADAPT_AWARE_SCH_SW** option of the **NRDUCellUlSch**.***UlSchAlgoSwitch*** parameter. The **NRDUCellUlSch**.***AdaptAwareSchTimeThld*** parameter specifies the duration threshold for adaptive awareness scheduling to take effect, and the **NRDUCellUlSch**.***AdaptAwareSchPriFactor*** parameter specifies the uplink data scheduling priority factor used in this function.

5.1.2.4 Uplink-Experience-based Scheduling Optimization

On a medium- or heavy-load live network with limited PUSCH scheduling resources, the primary objective of PUSCH resource allocation optimization should be optimizing uplink user experience. This is why uplink-experience-based scheduling optimization has been designed.

This function enables the gNodeB to flexibly adjust the uplink scheduling priorities of UEs based on UE types. Specifically:

- If a small-packet UE's retransmission scheduling duration is less than the threshold for delayed scheduling of uplink retransmission, the gNodeB adaptively adjusts the UE's retransmission scheduling priority to delay its scheduling.

- If a UE engaged in uplink preallocation has an average uplink MCS index less than the MCS threshold for preallocation period adjustment, the gNodeB adaptively adjusts the UE's preallocation period to speed up its scheduling.
- If a UE with low channel quality experiences consecutive bit errors, the gNodeB updates the UE's data information in a timely manner to save uplink scheduling resources.

This function is controlled by the **UL_EXP_SCH_OPT_SW** option of the **NRDUCellUlSch**.***UlSchAlgoSwitch*** parameter. The **NRDUCellUlSch**.***UlRetransSchDelayThld*** parameter specifies the threshold for delayed scheduling of uplink retransmission, and the **NRDUCellUlSch**.***PreSchPeriodAdjMcsThld*** parameter specifies the MCS index threshold for preallocation period adjustment.

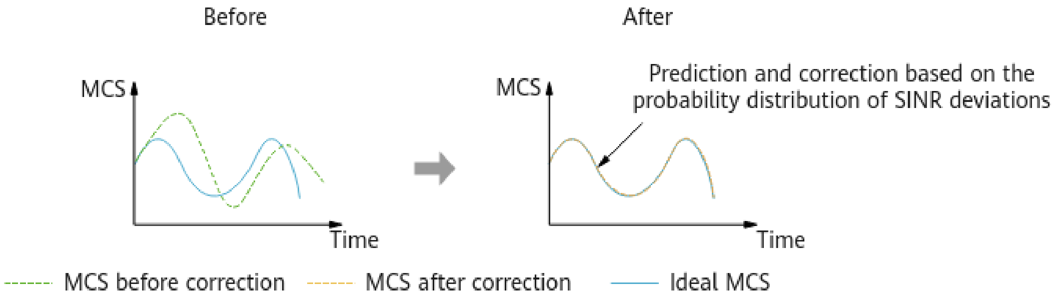
5.1.2.5 Probability-based Uplink MU-MIMO MCS Selection

On the live network, fluctuations in channel transmission quality affects signal transmission and the accuracy of uplink MCS selection for UEs involved in MU pairing. These impacts are even more significant under medium- or heavy-load conditions. To address this issue, probability-based uplink MU-MIMO MCS selection has been introduced.

During PUSCH MCS selection in MU-MIMO, this function estimates the extent of fluctuations in the channel transmission quality of UEs. It does so by analyzing the probability distribution of SINR deviations between the measurement moment and the scheduling moment. The estimation is then used to correct uplink MCS selection for UEs and thereby increase the average uplink UE throughput.

Figure 5-6 shows how this function works.

Figure 5-6 Probability-based uplink MU-MIMO MCS selection



This function is controlled by the **UL_MU_PBMCSSEL_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter.

5.1.3 Multi-Domain Demodulation for Interference Resistance

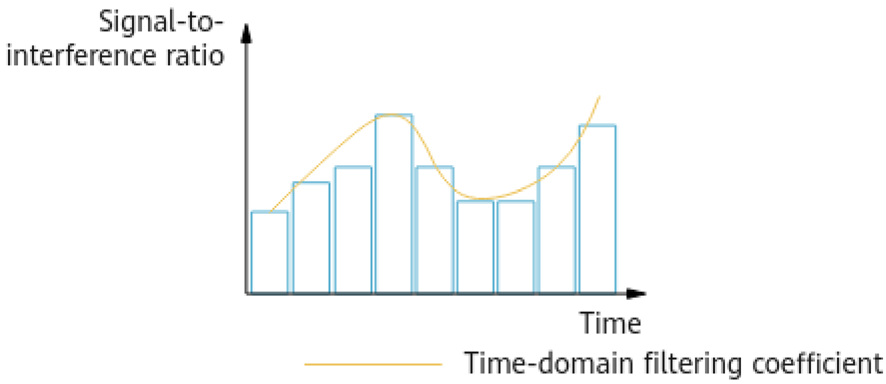
5.1.3.1 Multi-Beam Reception Enhancement

A higher number of receive beams boosts signal energy. However, it also leads to increased interference, compromising the accuracy of channel estimation. To further improve the accuracy of channel estimation in such scenarios, multi-beam reception enhancement has been introduced.

In a massive MIMO cell with multiple receive channels, **this function enables the gNodeB to adaptively adjust the time-domain filtering coefficient for the power delay profile (PDP) of PUSCH DMRS. This adjustment is based on the signal-to-interference ratio on receive beams, improving the accuracy of channel estimation and the uplink user-perceived rate.**

Figure 5-7 shows how this function works.

Figure 5-7 Multi-beam reception enhancement



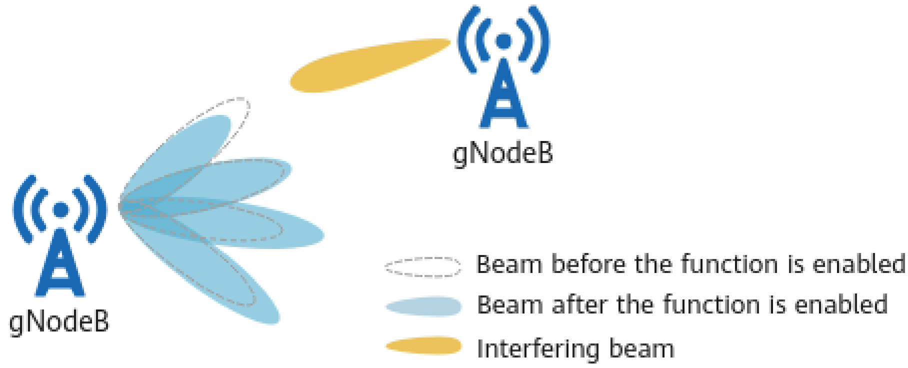
This function is controlled by the **MULTI_BEAM_RX_ENH_SW** option of the **NRDUCellPusch**.***PuschAlgoExtSwitch*** parameter.

5.1.3.2 Interference-based Precise Frequency Offset Estimation

Interference affects frequency offset estimation, which consequently introduces errors that result in inaccurate channel estimation. To address this, interference-based precise frequency offset estimation has been introduced to prevent inaccurate frequency offset estimation in the presence of interference, thereby improving channel estimation and uplink demodulation performance for UEs.

During uplink frequency offset estimation, this function enables the gNodeB to suppress interference in directions with strong interference and enhance wanted signals in directions with weak interference. This reduces the impact of interference on target beams, improving the accuracy of frequency offset estimation and uplink demodulation performance for UEs. Figure 5-8 shows how this function works.

Figure 5-8 Interference-based precise frequency offset estimation



This function is controlled by the **INTRF_PREC_FREQ_OFS_EST_SW** option of the **NRDUCellPusch**.***PuschMeasSw*** parameter.

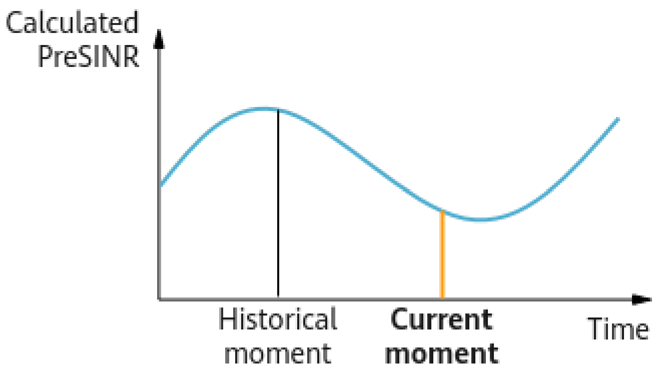
5.1.3.3 Real-Time-Information-based Precise Channel Estimation

Channel estimation based on historical PreSINRs is inaccurate and degrades demodulation performance as it fails to account for channel quality fluctuations in real time. Real-time-information-based precise channel estimation has been introduced to address this issue.

This function enables the gNodeB to leverage current information on scheduling and interference to calculate real-time PreSINRs for uplink channel estimation, thereby improving estimation accuracy.

Figure 5-9 shows how this function works.

Figure 5-9 Real-time-information-based precise channel estimation



This function is controlled by the **PREC_CHANNEL_EST_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter.

5.2 Network Analysis

5.2.1 Benefits

Uplink Boosting 2.0 enhances uplink cell performance through multi-cell power control for interference reduction, time-frequency scheduling for interference avoidance, and multi-domain demodulation for interference resistance, thereby increasing the average uplink UE throughput in medium- and heavy-load scenarios. This feature applies to cells providing contiguous coverage, and it delivers optimal benefits when all the following conditions are met:

- The uplink PRB usage of a cell is greater than or equal to 30%.
- The average uplink interference of neighboring cells (**N.UL.NI.Avg**) is greater than or equal to –108 dBm.
- The uplink CCE allocation success rate of a cell is less than or equal to 95% in the case that the uplink PRB usage of the cell is less than 50%.

The specific benefits depend on factors such as the traffic model and network load. Benefits are greater with a higher uplink PRB usage, higher average uplink interference, lower uplink CCE allocation success rate, and higher uplink MU pairing rate. When the cell load is light (for example, the uplink PRB usage is 20%) or the average uplink interference of neighboring cells (**N.UL.NI.Avg**) is less than –108 dBm, some sub-functions take effect in a smaller proportion of cases, due to which fewer or even no benefits will be obtained.

NOTE

Average uplink UE throughput = (**N.ThpVol.UL** – **N.ThpVol.UE.UL.SmallPkt**)/ **N.ThpTime.UE.UL.RmvSmallPkt**

5.2.2 Impacts

The impacts between Uplink Boosting 2.0 and other functions are described in **5.3.2.3 Function Impacts**.

The following impacts may be caused by all the functions of multi-cell power control for interference reduction, time-frequency scheduling for interference avoidance, and multi-domain demodulation for interference resistance:

- Due to improved uplink cell performance, the overall online duration of UEs is reduced, prolonging the DRX sleep time. This increases UEs' service delay and first packet delay, and consequently decreases the average downlink UE throughput. In addition, UEs performing periodic services will experience more re-synchronizations, potentially lowering their contention-based access success rate.
- In cells with only cell-center UEs or a large proportion of tail packet traffic, improved uplink cell performance may increase the uplink cell throughput and spectral efficiency while reducing the average uplink UE throughput.

Table 5-1 describes the impacts specific to each function on the network.

Table 5-1 Network impacts specific to each function

Function Type	Network Impact
Multi-cell power control for interference reduction	• Due to lower PUSCH transmit power in the serving cell, the following impacts will occur: – The RSRP and SINRs of SRSs and PUSCH in the cell may decrease. As a result, the average MCS index and IBLER in the uplink will change. In addition, more NR UEs will switch to LTE for uplink data transmission, increasing the value of the **N.NsaDc.ULChangetoMeNB** counter. – The uplink padding rate decreases. As a result, the uplink traffic volume and average uplink throughput at the MAC layer of the cell decrease, and the uplink PRB usage of the cell increases. – The uplink MU pairing rate changes. – The CPU usage may increase.
Time-frequency scheduling for interference avoidance	• As the retransmission priorities of UEs are adjusted in a cell, UEs may experience a few more retransmissions, and slightly higher uplink IBLER and RBLER. In addition, the uplink PRB usage and uplink MU pairing rate in the cell will change. • With UEs' scheduling priorities updated in a cell, the uplink PRB usage of the cell may change, and the average uplink throughput of cell-edge UEs may decrease. • The uplink padding rate in a cell decreases. As a result, the uplink traffic volume and average uplink throughput at the MAC layer of the cell decrease. • The average uplink MCS index may change. As a result, the uplink IBLER and RBLER may increase, and the uplink MU pairing rate may decrease, and consequently the uplink PRB usage decreases. • UE data is scheduled more quickly in a cell, reducing the number of BSRs. As a result, the average number of active UEs in the cell may decrease. • The CPU usage may increase.
Multi-domain demodulation for interference resistance	The average uplink MCS index, uplink IBLER, uplink RBLER, and PUSCH SINR in a cell may change.

Table 5-2 lists the performance indicators involved in the network impacts specific to each function.

Table 5-2 Performance indicators involved in the network impacts specific to each function

Indicator	Formula
DRX sleep time	**N.Cdrx.Sleep.Dur.Total**
Service delay	(**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num**)/1000 + (**N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**)/1000
First packet delay	(**N.Traffic.DL.RlcFirstPktDelay.Time**/ **N.Qos.DL.RlcFirstPktDelay.Num**)/1000
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average uplink UE throughput	**User Uplink Average Throughput (DU)**
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Average number of used PUSCH PRBs	**N.PRB.PUSCH.Used.Avg**
Number of uplink paired PRBs in a cell	**N.ChMeas.MIMO.UL.Pair.PRB**
Uplink RBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
Uplink PRB usage	**Uplink Resource Block Utilizing Rate (DU)**

Indicator	Formula
Average uplink MCS index	(0 x **N.ChMeas.PUSCH.MCS.0** + 1 x **N.ChMeas.PUSCH.MCS.1** + 2 x **N.ChMeas.PUSCH.MCS.2** + 3 x **N.ChMeas.PUSCH.MCS.3** + ... + 27 x **N.ChMeas.PUSCH.MCS.27** + 28 x **N.ChMeas.PUSCH.MCS.28**)/(**N.ChMeas.PUSCH.MCS.0** + **N.ChMeas.PUSCH.MCS.1** + **N.ChMeas.PUSCH.MCS.2** + **N.ChMeas.PUSCH.MCS.3** + ... + **N.ChMeas.PUSCH.MCS.27** + **N.ChMeas.PUSCH.MCS.28**)
Uplink MU pairing rate	**N.ChMeas.MIMO.UL.Pair.PRB**/(**N.PRB.PUSCH.Used.Avg** x **N.UL.PUSCH.Tti.Num**) x 100%
Average uplink interference	**N.UL.NI.Avg**
Average number of active UEs in a cell	**N.User.RRCConn.Active.Avg**
CPU usage	**VS.BBUBoard.CPULoad.Mean**

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-100201	Uplink Boosting 2.0	NR0S00UPBT20	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

Functions of multi-cell power control for interference reduction require the following functions.

- PUSCH coordinated power control

Function Name	Function Switch	Reference	Description
Intra-frequency neighboring cell measurement	**INTRA_FREQ_NCELL_STAT_SW** option of the **NRDUCellCollabServ**.***NeighborNrCellMgmtSw*** parameter	None	Enabling PUSCH coordinated power control requires that intra-frequency neighboring cell measurement be enabled.
SRS interference coordination based on self-contained slots	**SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	Enabling PUSCH coordinated power control requires that SRS interference coordination based on self-contained slots be enabled.
Closed-loop power control for the PUSCH in dynamic scheduling mode	**PUSCH_DYN_SCH_CLOSED_LOOP_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoSwitch*** parameter	*Power Control*	"Closed-loop power control for the PUSCH in dynamic scheduling mode" must be enabled for PUSCH coordinated power control to take effect.

- Link-performance-based power control

Function Name	Function Switch	Reference	Description
Closed-loop power control for the PUSCH in dynamic scheduling mode	**PUSCH_DYN_SCH_CLOSED_LOOP_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoSwitch*** parameter	*Power Control*	"Closed-loop power control for the PUSCH in dynamic scheduling mode" must be enabled for link-performance-based power control to take effect.

Functions of time-frequency scheduling for interference avoidance require the following functions.

- Uplink precise MCS optimization

Function Name	Function Switch	Reference	Description
Intelligent uplink multi-layer SINR prediction	**UL_SU_SINR_INTEL_PREDICT_SW** option of the **NRDUCellAiAlgo**.***AiAmcAlgoSwitch*** parameter	*MIMO (TDD)*	Intelligent uplink multi-layer SINR prediction must be enabled for uplink precise MCS optimization to take effect.
Additional DMRS	**NRDUCellPusch**.***UlAdditionalDmrsPos*** set to a value other than **NOT_CONFIG**	None	Uplink precise MCS optimization can be enabled only when **NRDUCellPusch**.***UlAdditionalDmrsPos*** is set to a value other than **NOT_CONFIG**.

- Uplink-experience-based scheduling optimization

Function Name	Function Switch	Reference	Description
SR-based scheduling optimization	**SR_SCH_OPT_SW** option of the **NRDUCellPusch**.***PuschPerformanceSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	SR-based scheduling optimization must be enabled for uplink-experience-based scheduling optimization to take effect.
Stopping uplink scheduling for uplink-abnormal UEs	**ABN_UE_STOP_ULSCH_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	*Uplink Scheduling*	"Stopping uplink scheduling for uplink-abnormal UEs" must be enabled for uplink-experience-based scheduling optimization to take effect.
Uplink basic preallocation	**UL_BASIC_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	Uplink-experience-based scheduling optimization can take effect only when uplink basic preallocation or uplink smart preallocation is enabled.

Function Name	Function Switch	Reference	Description
Uplink smart preallocation	**UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	

- Probability-based uplink MU-MIMO MCS selection

Function Name	Function Switch	Reference	Description
Probability-based uplink MCS selection	**UL_PBMCSSEL_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter	*Turbo Uplink (Low-Frequency TDD)*	Enabling probability-based uplink MU-MIMO MCS selection requires that probability-based uplink MCS selection be enabled.
PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	Probability-based uplink MU-MIMO MCS selection can take effect only when PUSCH MU-MIMO is enabled.

Functions of multi-domain demodulation for interference resistance require the following functions.

- Multi-beam reception enhancement

Function Name	Function Switch	Reference	Description
Time-domain adaptive filtering	**PUSCH_TIME_DOM_ADAPT_FILTER_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	Time-domain adaptive filtering must be enabled for multi-beam reception enhancement to take effect.

- Interference-based precise frequency offset estimation

Function Name	Function Switch	Reference	Description
Uplink DMRS Type	**NRDUCellPusch**.***UlDmrsType***	None	Interference-based precise frequency offset estimation requires that the **NRDUCellPusch**.***UlDmrsType*** parameter be set to **TYPE1**.

5.3.2.2 Mutually Exclusive Functions

Uplink Boosting 2.0 has the following mutually exclusive functions.

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	High-speed cells do not support Uplink Boosting 2.0.

Functions of multi-cell power control for interference reduction have the following mutually exclusive functions.

PUSCH coordinated power control

Function Name	Function Switch	Reference	Description
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Hyper cells do not support PUSCH coordinated power control.
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** or **DPS_MODE**	*Cell Combination*	For an NR TDD combined cell in basic or dynamic point selection (DPS) transmission mode, PUSCH coordinated power control cannot be enabled.

Function Name	Function Switch	Reference	Description
Joint Transmission Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** and **NRDUCellMultiTrp**.***DataTransMode*** set to **JOINT_MODE**	*Cell Combination*	For an NR TDD cell with Joint Transmission Cell Combination enabled, PUSCH coordinated power control cannot be enabled.
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	PUSCH coordinated power control and Fusion Cell cannot be both enabled.
Positioning-dedicated SRS transmission in uplink slots	**SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter	*Positioning*	PUSCH coordinated power control and positioning-dedicated SRS transmission in uplink slots cannot be both enabled.

Functions of time-frequency scheduling for interference avoidance have the following mutually exclusive functions.

Uplink precise MCS optimization

Function Name	Function Switch	Reference	Description
Uplink scheduling on a per RB basis	**UL_1RB_SCH_SWITCH** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	*Uplink Scheduling*	Uplink scheduling on a per RB basis and uplink precise MCS optimization cannot be both enabled.

Functions of multi-domain demodulation for interference resistance have the following mutually exclusive functions.

Real-time-information-based precise channel estimation

Function Name	Function Switch	Reference	Description
3D communication coverage mode	**NRDUCell**.***TridCommMode*** set to **TRID_COVERAGE_MODE** and **NRDUCell**.***AerialCellFlag*** set to **GROUND_AERIAL_CELL**	None	Real-time-information-based precise channel estimation cannot be enabled if both of the following conditions are met: ● The **NRDUCell**.***TridCommMode*** parameter is set to **TRID_COVERAGE_MODE**. ● The **NRDUCell**.***AerialCellFlag*** parameter is set to **GROUND_AERIAL_CELL**.

5.3.2.3 Function Impacts

Functions of time-frequency scheduling for interference avoidance have the following function impacts.

- Uplink precise MCS optimization

Function Name	Function Switch	Reference	Description
Uplink PAMC	**UL_PAMC_SWITCH** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	*Uplink Scheduling*	Uplink precise MCS optimization preferentially takes effect when it is enabled together with uplink PAMC.
Uplink target IBLER adaptation	**UL_IBLER_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter	*Uplink Scheduling*	Uplink precise MCS optimization preferentially takes effect when it is enabled together with uplink target IBLER adaptation.

Function Name	Function Switch	Reference	Description
Closed-loop PUSCH antenna selection	**PUSCH_CLOSED_LOOP_ANT_SEL_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	Closed-loop PUSCH antenna selection preferentially takes effect when it is enabled together with uplink precise MCS optimization.
Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	Uplink precise MCS optimization does not take effect for UEs with VoNR in effect.
Identification of UEs requiring connectivity with low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	Uplink precise MCS optimization does not take effect for UEs with the low latency and high reliability function in effect.
Probability-based uplink MCS selection	**UL_PBMCSSEL_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter	*Turbo Uplink (Low-Frequency TDD)*	Uplink precise MCS optimization preferentially takes effect when it is enabled together with probability-based uplink MCS selection.
Adaptive outer-loop adjustment for small-packet services	**SMALL_PKT_OL_ADAPT_ADJ_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	Uplink precise MCS optimization preferentially takes effect when it is enabled together with "adaptive outer-loop adjustment for small-packet services".
Uplink 4-layer SU transmission	**UL_SU_FOUR_LAYER_SW** option of the **NRDUCellUlRank**.***UlRankAlgoSw*** parameter	None	Uplink precise MCS optimization does not take effect for UEs with rank 3 or rank 4 in the uplink.

- Adaptive awareness scheduling

Function Name	Function Switch	Reference	Description
Packet-length-based scheduling optimization	**PKT_LEN_BASED_SCH_OPT_SW** option of the **NRDUCellPusch**.***PuschPerformanceSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	Packet-length-based scheduling optimization does not take effect when it is enabled together with adaptive awareness scheduling.

- Probability-based uplink MU-MIMO MCS selection

Function Name	Function Switch	Reference	Description
Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	Probability-based uplink MU-MIMO MCS selection does not take effect for UEs with VoNR in effect.
Identification of UEs requiring connectivity with low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	Probability-based uplink MU-MIMO MCS selection does not take effect for UEs with the low latency and high reliability function in effect.
Uplink PAMC	**UL_PAMC_SWITCH** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	*Uplink Scheduling*	Probability-based uplink MU-MIMO MCS selection preferentially takes effect when it is enabled together with uplink PAMC.
Uplink target IBLER adaptation	**UL_IBLER_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter	*Uplink Scheduling*	Probability-based uplink MU-MIMO MCS selection preferentially takes effect when it is enabled together with uplink target IBLER adaptation.

Functions of multi-domain demodulation for interference resistance have the following function impacts.

- Multi-beam reception enhancement

Function Name	Function Switch	Reference	Description
PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	Multi-beam reception enhancement does not take effect for UEs with PUSCH MU-MIMO in effect.
Identification of UEs requiring connectivity with low latency and high reliability	**gNBQciBearer**.***ServiceType*** set to **URLLC_HIGH_RELIABILITY_SERVICE**	*URLLC*	Multi-beam reception enhancement does not take effect for UEs requiring connectivity with low latency and high reliability.

- Interference-based precise frequency offset estimation

Function Name	Function Switch	Reference	Description
Frequency offset measurement optimization based on interference suppression	**INTRF_FREQOFFSET_OPT_SW** option of the **NRDUCellPusch**.***PuschMeasSw*** parameter	*Interference Avoidance*	Frequency offset measurement optimization based on interference suppression does not take effect when it is enabled together with interference-based precise frequency offset estimation.

- Real-time-information-based precise channel estimation

Function Name	Function Switch	Reference	Description
PUSCH beam domain enhancement	**PUSCH_BEAM_DOMAIN_ENH_SW** option of the **NRDUCellAlgoSwitch**.***FullChannelCovEnhSwitch*** parameter	*MIMO (TDD)*	Real-time-information-based precise channel estimation does not take effect when it is enabled together with PUSCH beam domain enhancement.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

PUSCH coordinated power control, uplink precise MCS optimization, probability-based uplink MU-MIMO MCS selection, link-performance-based power control, and multi-beam reception enhancement:

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. Uplink precise MCS optimization requires that 5900 series base stations be not configured with the BBU5900C or BookBBU5901.

Other functions:

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

If the main control board is UMPTe, uplink precise MCS optimization is supported only in NR-only scenarios. Table 5-3 lists the uplink precise MCS optimization model specifications supported by each type of main control board working in different modes.

Table 5-3 Model specifications

Main Control Board	Working Mode	Model Specifications
UMPTg and later	NR-only	36
	Co-MPT multiple modes	18
UMPTga	NR-only	18
	Co-MPT multiple modes	9
UMPTe	NR-only	18
	Co-MPT multiple modes	0

The other functions are supported by all NR-capable main control boards and NR TDD-capable baseband processing units. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Real-time-information-based precise channel estimation and interference-based precise frequency offset estimation: All NR TDD-capable 32T32R and 64T64R RF modules that work in low frequency bands can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Other functions: All NR TDD-capable low-frequency RF modules with at least 32T32R can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Networking

All the following conditions must be met for PUSCH coordinated power control to take effect:

- eXn interfaces must be set up between base stations.
- The inter-base-station transmission delay must be less than or equal to 4 ms. The inter-base-station transmission delay (unit: 0.1 microsecond) equals max(One-way delay, Two-way delay minus one-way delay).
  - One-way delay refers to the one-way delay on the user plane of the gNodeB DU eXn interface. It is indicated by the **VS.IPPM.Owd.Means** counter.
  - Two-way delay refers to the two-way delay on the user plane of the gNodeB DU eXn interface. It is indicated by the **VS.IPPM.HighLevel.Rtt.Means** counter.
- The base stations must be time-synchronized to within 1.5 µs. To check the clock quality, run the **STR CLKTST** command to start clock quality monitoring and then run the **DSP CLKTST** command to view the monitoring results.
- The bandwidths of the links between the base stations must meet certain requirements.

  The required link bandwidth can be planned based on network conditions. It must be within the transmission bandwidth capacities of the base stations.

For details about how to configure an eXn interface and how to check whether an eXn interface has been successfully set up, see *eXn Self-Management*.

The other functions have no requirements.

5.3.5 Cells

Uplink Boosting 2.0 has the following cell requirements:

- The **NRDUCell**.***DuplexMode*** parameter is set to **CELL_TDD**.
- The **NRDUCell**.***FrequencyBand*** parameter is set to **N38**, **N40**, **N41**, **N77**, **N78**, or **N79**.
- When the **NRDUCell**.***LampSiteCellFlag*** parameter is set to **YES**, the **NRDUCellTrp**.***TrpType*** parameter is set to **MASTER_DM_MIMO** or **MASTER_DM_MIMO_LIGHT**.
- (Recommended) The **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **NORMAL_CELL**.

PUSCH coordinated power control has the following requirements:

- The **NRDUCell**.***SlotAssignment*** parameter is set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**.
- The **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **NORMAL_CELL**.

5.3.6 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-4 describes the parameters used for activating Uplink Boosting 2.0.

Table 5-4 Parameters used for activating Uplink Boosting 2.0

Parameter Name	Parameter ID	Option	Setting Notes
MIMO Feature Switch	**NRDUCellFeatureSw**.***MimoFeatureSwitch***	UL_LOW_NOISE_PHASE2_SW	Select this option if Uplink Boosting 2.0 is required.

Table 5-5 describes the parameters used for activating functions of multi-cell power control for interference reduction.

Table 5-5 Parameters used for activating functions of multi-cell power control for interference reduction

Parameter Name	Parameter ID	Option	Setting Notes
UL Power Control Algorithm Extension Switch	**NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch***	PUSCH_COORD_PWR_CTRL_SW	Select this option if PUSCH coordinated power control is required.
UL Coordinated Power Control UE SSB RSRP Thld	**NRCellMeasConfig**.***UlCpcUeSsbRsrpThld***	None	Set this parameter based on the network plan if PUSCH coordinated power control is required.
PUSCH Joint Power Control Cell IoT Thld	**NRDUCellUlPcConfig**.***PuschJointPcCellIotThld***	None	Set this parameter based on the network plan if PUSCH coordinated power control is required.

Parameter Name	Parameter ID	Option	Setting Notes
UL Coordinated Power Control UE SRS RSRP Thld	**NRDUCellUlPcConfig**.***UlCpcUeSrsRsrpThld***	None	Set this parameter based on the network plan if PUSCH coordinated power control is required.
UL Coordinated Power Control UE Intrf RSRP Thld	**NRDUCellUlPcConfig**.***UlCpcUeIntrfRsrpThld***	None	Set this parameter based on the network plan if PUSCH coordinated power control is required.
PUSCH Coordinated Power Control PRB Threshold	**NRDUCellUlPcConfig**.***PuschCoordPcPrbThld***	None	Set this parameter based on the network plan if PUSCH coordinated power control is required.
UL Power Control Algorithm Extension Switch	**NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch***	INTRF_SC_LINK_PERF_PC_SW	Select this option if link-performance-based power control is required.
Link Performance PC RSRP Thld Offset	**NRDUCellUlPcConfig**.***LinkPerfPcRsrpThldOffset***	None	Set this parameter based on the network plan if link-performance-based power control is required.

Table 5-6 describes the parameters used for activating functions of time-frequency scheduling for interference avoidance.

Table 5-6 Parameters used for activating functions of time-frequency scheduling for interference avoidance

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Algorithm Extension Switch	**NRDUCellPusch**.***PuschAlgoExtSwitch***	UL_RETRANS_PREC_RB_CTRL_SW	Select this option if precise RB control for uplink retransmissions is required.

Parameter Name	Parameter ID	Option	Setting Notes
UL First Retrans Minimum RB Percentage	**NRDUCellUlSch**.***UlFirstRetransMinRbPct***	None	It is recommended that the **NRDUCellUlSch**.***UlFirstRetransMinRbPct*** parameter be set to **20** when precise RB control for uplink retransmissions is enabled, to ensure optimal network performance.
AI AMC Algorithm Switch	**NRDUCellAiAlgo**.***AiAmcAlgoSwitch***	UL_PRECISE_MCS_OPT_SW	Select this option if uplink precise MCS optimization is required.
UL MCS Predict Error Upper Limit	**NRDUCellAiAlgo**.***UlMcsPredictErrUpLimit***	None	Set this parameter based on the network plan if uplink precise MCS optimization is required.
UL Scheduling Algorithm Switch	**NRDUCellUlSch**.***UlSchAlgoSwitch***	ADAPT_AWARE_SCH_SW	Select this option if adaptive awareness scheduling is required.
Adaptive Aware Scheduling Time Threshold	**NRDUCellUlSch**.***AdaptAwareSchTimeThld***	None	Set this parameter based on the network plan if adaptive awareness scheduling is required.
Adaptive Aware Scheduling Priority Factor	**NRDUCellUlSch**.***AdaptAwareSchPriFactor***	None	Set this parameter based on the network plan if adaptive awareness scheduling is required.
UL Scheduling Algorithm Switch	**NRDUCellUlSch**.***UlSchAlgoSwitch***	UL_EXP_SCH_OPT_SW	Select this option if uplink-experience-based scheduling optimization is required.
UL Retransmission Scheduling Delay Threshold	**NRDUCellUlSch**.***UlRetransSchDelayThld***	None	Set this parameter based on the network plan if uplink-experience-based scheduling optimization is required.
Pre Scheduling Period Adjust MCS Threshold	**NRDUCellUlSch**.***PreSchPeriodAdjMcsThld***	None	Set this parameter based on the network plan if uplink-experience-based scheduling optimization is required.
Uplink AMC Performance Switch	**NRDUCellUlAmc**.***UlAmcPerformanceSw***	UL_MU_PBMCSSEL_SW	Select this option if probability-based uplink MU-MIMO MCS selection is required.

Table 5-7 describes the parameters used for activating functions of multi-domain demodulation for interference resistance.

Table 5-7 Parameters used for activating functions of multi-domain demodulation for interference resistance

Parameter Name	Parameter ID	Option	Setting Notes
PUSCH Algorithm Extension Switch	**NRDUCellPusch.*PuschAlgoExtSwitch***	MULTI_BEAM_RX_ENH_SW	Select this option if multi-beam reception enhancement is required.
PUSCH Measurement Sw	**NRDUCellPusch.*PuschMeasSw***	INTRF_PREC_FREQ_OFS_EST_SW	Select this option if interference-based precise frequency offset estimation is required.
Uplink PUSCH Algorithm Switch	**NRDUCellPusch.*UlPuschAlgoSwitch***	PREC_CHANNEL_EST_SW	Select this option if real-time-information-based precise channel estimation is required.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Enabling Uplink Boosting 2.0

```
//Turning on the switch for Uplink Boosting 2.0
MOD NRDUCELLFEATURESW: NrDuCellId=0, MimoFeatureSwitch=UL_LOW_NOISE_PHASE2_SW-1;
```

Enabling functions of multi-cell power control for interference reduction

```
//Enabling PUSCH coordinated power control and setting the UlCpcUeSsbRsrpThld, UlCpcUeSrsRsrpThld, UlCpcUeIntrfRsrpThld, PuschJointPcCellIotThld, and PuschCoordPcPrbThld parameters
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoExtSwitch=PUSCH_COORD_PWR_CTRL_SW-1;
MOD NRCELLMEASCONFIG: NrCellId=0, UlCpcUeSsbRsrpThld=-120;
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlCpcUeSrsRsrpThld=-98, UlCpcUeIntrfRsrpThld=-110, PuschJointPcCellIotThld=5, PuschCoordPcPrbThld=30;
//Enabling link-performance-based power control and setting the LinkPerfPcRsrpThldOffset parameter
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoExtSwitch=INTRF_SC_LINK_PERF_PC_SW-1, LinkPerfPcRsrpThldOffset=0;
```

Enabling functions of time-frequency scheduling for interference avoidance

```
//Enabling precise RB control for uplink retransmissions
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=UL_RETRANS_PREC_RB_CTRL_SW-1;
MOD NRDUCELLULSCH: NrDuCellId=0, UlFirstRetransMinRbPct=20;
//Enabling uplink precise MCS optimization
MOD NRDUCELLAIALGO: NrDuCellId=0, AiAmcAlgoSwitch=UL_PRECISE_MCS_OPT_SW-1,
UlMcsPredictErrUpLimit=5;
//Enabling adaptive awareness scheduling
MOD NRDUCELLULSCH: NrDuCellId=0, UlSchAlgoSwitch=ADAPT_AWARE_SCH_SW-1,
AdaptAwareSchPriFactor=500, AdaptAwareSchTimeThld=0;
//Enabling uplink-experience-based scheduling optimization and setting the UlRetransSchDelayThld and
PreSchPeriodAdjMcsThld parameters
MOD NRDUCELLULSCH: NrDuCellId=0, UlSchAlgoSwitch=UL_EXP_SCH_OPT_SW-1,
UlRetransSchDelayThld=12, PreSchPeriodAdjMcsThld=17;
//Enabling probability-based uplink MU-MIMO MCS selection
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcPerformanceSw=UL_MU_PBMCSSEL_SW-1;
```

Enabling functions of multi-domain demodulation for interference resistance

```
//Enabling multi-beam reception enhancement
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=MULTI_BEAM_RX_ENH_SW-1;
//Enabling interference-based precise frequency offset estimation
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=INTRF_PREC_FREQ_OFS_EST_SW-1;
//Enabling real-time-information-based precise channel estimation
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=PREC_CHANNEL_EST_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Disabling functions of multi-cell power control for interference reduction

```
//Disabling link-performance-based power control
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoExtSwitch=INTRF_SC_LINK_PERF_PC_SW-0;
//Disabling PUSCH coordinated power control
MOD NRDUCELLULPCCONFIG: NrDuCellId=0, UlPwrCtrlAlgoExtSwitch=PUSCH_COORD_PWR_CTRL_SW-0;
```

Disabling functions of time-frequency scheduling for interference avoidance

```
//Disabling probability-based uplink MU-MIMO MCS selection
MOD NRDUCELLULAMC: NrDuCellId=0, UlAmcPerformanceSw=UL_MU_PBMCSSEL_SW-0;
//Disabling uplink-experience-based scheduling optimization
MOD NRDUCELLULSCH: NrDuCellId=0, UlSchAlgoSwitch=UL_EXP_SCH_OPT_SW-0;
//Disabling adaptive awareness scheduling
MOD NRDUCELLULSCH: NrDuCellId=0, UlSchAlgoSwitch=ADAPT_AWARE_SCH_SW-0;
//Disabling uplink precise MCS optimization
MOD NRDUCELLAIALGO: NrDuCellId=0, AiAmcAlgoSwitch=UL_PRECISE_MCS_OPT_SW-0;
//Disabling precise RB control for uplink retransmissions
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=UL_RETRANS_PREC_RB_CTRL_SW-0;
```

Disabling functions of multi-domain demodulation for interference resistance

```
//Disabling real-time-information-based precise channel estimation
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPuschAlgoSwitch=PREC_CHANNEL_EST_SW-0;
//Disabling interference-based precise frequency offset estimation
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschMeasSw=INTRF_PREC_FREQ_OFS_EST_SW-0;
//Disabling multi-beam reception enhancement
MOD NRDUCELLPUSCH: NrDuCellId=0, PuschAlgoExtSwitch=MULTI_BEAM_RX_ENH_SW-0;
```

Disabling Uplink Boosting 2.0

```
//Turning off the switch for Uplink Boosting 2.0
MOD NRDUCELLFEATURESW: NrDuCellId=0, MimoFeatureSwitch=UL_LOW_NOISE_PHASE2_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

After the functions of Uplink Boosting 2.0 are enabled for a cell, observe related performance indicators to check whether the functions have taken effect. Specifically:

Multi-cell power control for interference reduction has taken effect if the cell's average uplink interference decreases or the average uplink UE throughput increases.

Time-frequency scheduling for interference avoidance has taken effect if the average uplink UE throughput increases.

Multi-domain demodulation for interference resistance has taken effect if the average uplink UE throughput increases.

To verify whether uplink precise MCS optimization has taken effect, you can also run the **DSP NRDUCELLAISCHMODEL** command to query model usage. If the **AI Case** is **UL Precise MCS Optimization** and the corresponding **AI Model Status** is **Building**, **In Use**, or **Updating**, uplink precise MCS optimization has taken effect.

5.4.3 Network Monitoring

After the functions of Uplink Boosting 2.0 are enabled, use the performance indicators described in **5.2 Network Analysis** to monitor cell performance.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

1. *Channel Management*
2. *MIMO (TDD)*
3. *High Speed Mobility*
4. *Cell Combination*
5. *Distributed Antenna Solutions (Low-Frequency TDD)*
6. *CA Experience Improvement*
7. *Power Control*
8. *License Control Item Lists*
9. *NR Inter-Carrier Dynamic Power Allocation*
10. *Turbo Uplink (Low-Frequency TDD)*
11. *RedCap*
12. *UE Power Saving*
13. *Coordinated Interference Management (Low-Frequency TDD)*
14. *Multi-Frequency Smart Selection*
15. *CoMP*
16. *Cell Management*
17. *Super Uplink*
18. *LTE and NR Zero Bufferzone*
19. *Uplink Scheduling*
20. *Stable Video Surveillance Experience*
21. *DRX*
22. *Energy Conservation and Emission Reduction*
23. *Positioning*
24. *Interference Avoidance*
25. *URLLC*
26. *Hyper Cell*
27. *eXn Self-Management*
28. *VoNR*

29. *Fusion Cell (Low-Frequency TDD)*